

The Arrow of Time as Standing Monotonicity
AASC Time Non-Primitivity I: Direction without Primitive Temporal Flow
Directionality, Records, Entropy, Causality, and Cosmological Asymmetry under AASC

Amos Jay Maley

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Abstract

This is the first paper in the AASC time non-primitivity sequence. It treats temporal direction rather than physical clock choice. The AASC kernel is instantiated by necessity in every non-degenerate regime that supports admissibility-assessable, identity-preserving standing-bearing construction. It is not generated by the arrow theorem or by entropy, causality, metric time, clocks, probability,

information, or cosmology. This manuscript begins after kernel closure and asks what temporal directionality is forced in regimes that remain kernel-faithful.

The central claim is that the invariant arrow is standing monotonicity in quotient form, not primitive temporal flow, entropy, causality, probability, information, memory, quantum collapse, or cosmological initial condition. AASC is a constraint formalism. Its primary operational mode is exhaustion and impossibility: once the target and kernel-faithful comparison class are fixed, every same-scope primitive-arrow route must preserve the required standing-bearing roles or resolve as presentation skin, anchor motion, hidden tensor load, scope change, repair, representative privilege, effective projection, or failure. The theorem object is closure, not a rival temporal substance, replacement entropy, preferred foliation, collapse mechanism, cosmological primitive, or information ontology.

The positive result is standing monotonicity in quotient form. Admissible irreversible fixation induces a standing preorder on raw standing-bearing stages. Quotienting by standing equivalence yields a partial order on standing-equivalence classes, and the standing arrow is the strict non-invertible direction of that quotient order. The paper therefore does not claim a partial order on raw stages.

Primitive temporal direction, primitive entropy, primitive information, primitive probability, primitive causality, psychological memory, quantum measurement selection, and privileged cosmological boundary conditions are inadmissible as foundations of standing-bearing directionality because each presupposes the standing discipline needed to identify, compare, transport, or reuse its own content. Physical time is retained-consequence ordering inside the realized physical interior; causality is witnessed dependence across that order; records are fixed witnesses available for downstream admissible reuse. Thermodynamic, record, causal, quantum-measurement, psychological, and cosmological arrows survive as effective projections or refinements. They may align locally without being identical in principle or required to align globally.

The result is a unified constraint-formalism account in which familiar arrows survive as admissible, plural, and non-primitive projections or refinements over standing monotonicity. Because the quotient order is partial rather than globally linear, the standing arrow supplies an invariant orientation wherever standing comparison is fixed while allowing incomparable branches unless a domain-specific projection supplies additional comparability.

Roadmap for the reader

This paper addresses a familiar puzzle in a different order from most arrow-of-time accounts. Standard discussions often ask whether the arrow is grounded in entropy, causality, records, cosmological initial conditions, quantum measurement, or psychological memory. AASC asks a prior question: what must already be in place for any of those candidate arrows to be a stable, reusable, standing-bearing claim rather than a presentation-dependent label?

The dependency order is fixed: kernel by necessity; downstream consequence by kernel-faithful specialization. Non-degenerate standing-bearing construction already instantiates admissibility, standing, reference, and irreversibility. The present paper neither chooses nor generates that kernel. It proves the directionality structure forced when a realized physical regime preserves it under lawful redescription and no same-act repair.

The answer is **standing monotonicity**. When an admissible act fixes standing, later admissible use may inherit, condition on, transform, or enrich that fixation, but it may not erase the fact of

fixation as the same act. On raw stages this gives a preorder. After quotienting stages that are mutually standing-equivalent, it gives a partial order. The strict direction of that quotient order is the standing arrow.

This differs from entropy-first accounts in practice. Entropy remains physically important, but it is not asked to ground directionality before state spaces, macrostates, measures, coarse-grainings, and record ledgers have standing. It differs from causality-first accounts because causal dependence is treated as a witnessed refinement over an already admitted order rather than as primitive influence. It differs from information-first accounts because information measures require encoding, reference, and redescription-stable comparison. It differs from cosmological-boundary accounts because low-entropy initial conditions may be declared model anchors, but they cannot authorize the admissibility conditions needed to identify them as physically privileged.

The intended audience is readers in foundations of physics, philosophy of physics, mathematical physics, and constraint-based formal theory. No prior commitment to the AASC corpus is assumed. The paper uses a compact AASC vocabulary, but every central term is either defined in the body or summarized in Appendix C. Readers mainly interested in the physical consequences can begin with the core disambiguation table, the entropy interface, the quantum-record section, the cosmological taxonomy, and the three case studies in Part VI.

The paper’s main message can be stated without notation:

The familiar arrows of time are real as effective arrows, but none is admissible as a primitive ground of directionality. What is forced is the non-invertible orientation of standing fixation; entropy, records, causality, quantum measurement, memory, and cosmology become domain-specific projections or refinements over that orientation.

The theorem target is the dependency and closure structure that any physical arrow explanation must preserve. Numerical entropy production, derivation of cosmological boundary conditions, decoherence dynamics, and complete cosmological modeling remain downstream sectoral tasks. Their separation from the present theorem is a scope distinction, not a qualification of the quotient conclusion.

Series position in the AASC time non-primitivity sequence

This manuscript is Paper I of a two-paper time non-primitivity sequence. Its target is temporal direction: what remains of the arrow of time once primitive temporal flow, primitive metric time, primitive entropy, primitive information, primitive probability, primitive causality, and primitive cosmological boundary authority are disallowed. Paper II, *The Physical Clock as a Quotient* (DOI: 10.5281/zenodo.20833048), treats the distinct clock-carrier problem: when a variable, subsystem, matter field, modular flow, Page–Wootters conditioning, semiclassical parameter, or record sequence has standing as a physical clock for a declared use. The Clock choice is assigned to Paper II, and the clock paper is not a premise of the present theorem. Paper I supplies the directionality and no-backflow discipline that Paper II specializes.

The sequence therefore separates two questions that are often collapsed. Paper I asks what fixes temporal orientation. Paper II asks what counts as a certified carrier for temporal readout, ordering, conditioning, or synchronization. The shared dependency lock is:

Clock standing is not temporal direction; temporal direction is not clock selection.

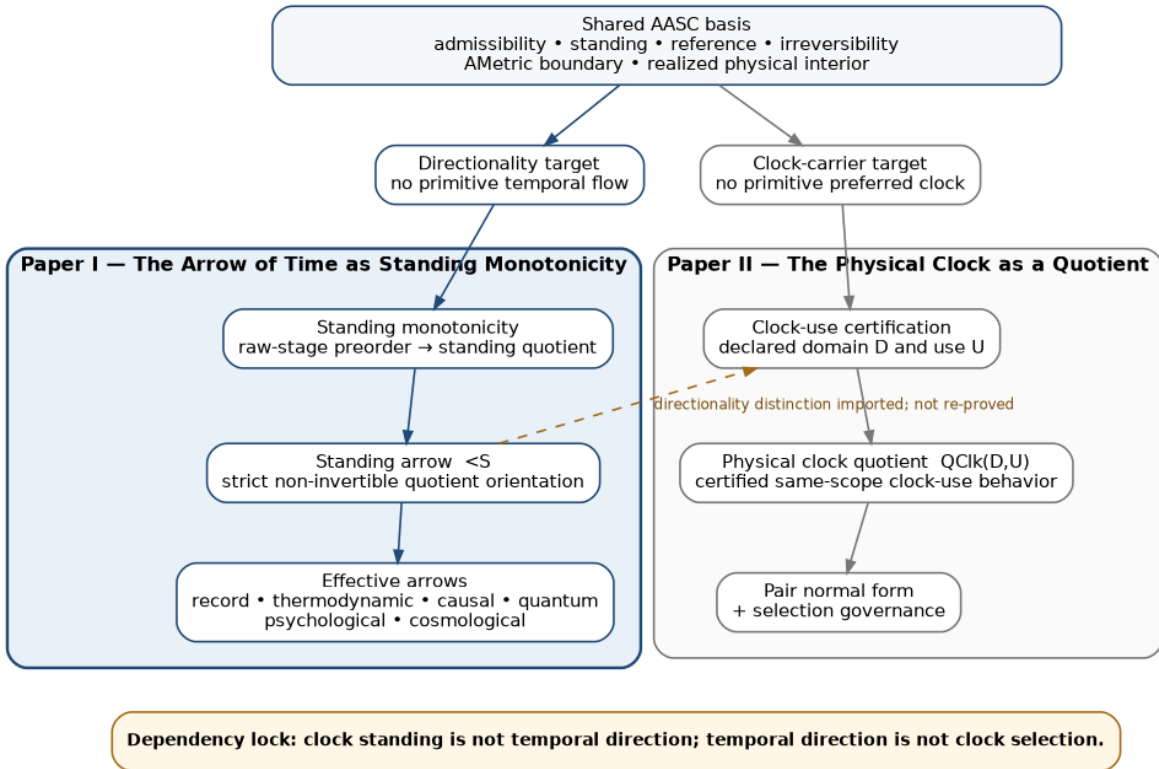


Figure 1: Architecture of the AASC time non-primitivity sequence, with Paper I emphasized. The clock sequel imports the directionality distinction established here but is not used to prove the standing arrow.

AASC constraint-formalism method

AASC is a constraint formalism. Its primary operational mode is exhaustion and impossibility rather than the construction of a rival mechanism, preferred representative, hidden substrate, or substitute dynamics. Once a fixed target is declared and the non-degenerate kernel is instantiated, AASC asks which primitive sources, selectors, repairs, transfers, representatives, or independent authorities remain available without changing scope or breaking kernel faithfulness.

A constructive mechanism program ordinarily asks which entropy law, causal primitive, clock, collapse rule, or cosmological boundary should be selected. The AASC question is prior and closure-forming: after the same-scope routes are exhausted, what invariant structure survives? In the present target, the surviving structure is standing monotonicity in quotient form. The eliminative route is therefore the positive identification method; the survivor is not chosen by preference but forced by the failure of every governance-non-equivalent same-scope alternative.

This exhaustion-first posture is the source of the formalism's novelty and force. A responsive objection must preserve the theorem class: it must deny target instantiation, break an upstream necessity theorem, identify an invalid specialization, or exhibit a kernel-faithful same-regime survivor outside the stated normal forms. A request for a more familiar constructive mechanism or preferred representative is not a refutation unless it supplies such a survivor.

Standing of objections. A criticism has standing only if it preserves the fixed target and kernel-faithful comparison class and then identifies an invalid specialization, an incomplete exhaustion branch, or a same-scope survivor. A request for a familiar mechanism, preferred representative, softer methodology, or conventional explanatory primitive does not challenge the theorem unless it supplies one of those witnesses.

Part I

Boundary and Dependency Discipline

1 Reader's guide and claim-status discipline

This paper is a fixed-domain constraint theorem for temporal directionality. It operates upstream of mechanism selection and model construction. Its target is the invariant directionality forced once primitive time-flow, primitive metric time, primitive probability, primitive information, primitive entropy, primitive causality, and privileged initial conditions are denied boundary-authorizing status in a kernel-faithful regime.

The paper has three arcs.

1. The necessity arc eliminates primitive arrow sources and proves that standing monotonicity is forced.
2. The projection arc classifies records, entropy, information, causality, quantum measurement, and psychological time as downstream structures over admissible standing.
3. The physics-facing arc explains why thermodynamic, quantum-record, relativistic, and cosmological arrows survive as effective arrows without becoming primitive foundations.

The dependency order is locked. The AASC kernel is not an optional formalism imported after the physical discussion begins. It is instantiated by necessity wherever a non-degenerate regime treats its outputs, records, states, or continuations as admissibility-assessable, identity-preserving standing-bearing content. The AMetric/interface discipline then blocks primitive metric, coordinate, scale, representative, probabilistic, and skin-as-load authority at the boundary. The realized-physical-interior arc supplies locality, comparison, transport, fields, particles, gauge, mass, vacuum, time, causality, and records as downstream physical roles. This paper does not use standard physics to authorize the kernel; it specializes the kernel and realized-interior architecture to classify directionality and the familiar arrows.

Kernel-faithful regime. A regime is kernel-faithful when its standing-bearing claims preserve admissibility, standing, reference, and irreversibility; preserve the distinction between admissible continuation and prohibited same-act repair; preserve target identity under lawful redescription; and introduce no metric, probability, typicality rule, selector, representative, clock, entropy scalar, causal primitive, cosmological boundary, or information measure as boundary-authoritative before admission.

Layer	Status in this paper	Function
AASC kernel	Necessity-instantiated upstream theorem family	In every non-degenerate regime supporting admissibility-assessable, identity-preserving construction, admissibility, standing, reference, and irreversibility are already in force. This paper does not generate them; it proves their downstream directionality consequence.
AMetric/interface discipline	Imported theorem family	Blocks primitive metric, coordinate, temporal, probabilistic, or representative authority before admission.
Realized physical interior	Imported role architecture	Supplies the physical ladder through non-null realization, localization, comparison, transport, field, particle, gauge, mass, vacuum, time, causality, and records.
Standing arrow	Proved here	The strict invariant directionality on standing-equivalence classes induced by irreversible standing fixation.
Record arrow	Proved here using imported record quotient	Explains past-fixation, evidence, memory, and record reuse.
Thermodynamic arrow	Derived effective projection	Classifies entropy increase as an admitted coarse-graining monotone, not a primitive arrow.
Causal arrow	Derived refinement	Classifies causality as witnessed dependence across standing quotient order, not as primitive influence.
Quantum measurement arrow	Classified	Places measurement direction in record fixation and quotient standing, not primitive collapse or consciousness.
Cosmological arrow	Classified	Places cosmological asymmetry in witness availability, constraint reachability, and declared boundary anchors.
Full thermodynamics, full cosmology, full quantum gravity	Theorem boundary	Downstream research programs requiring sector-specific witnesses and scope declarations.

Central thesis. AASC does not eliminate the arrow of time. It eliminates primitive temporal direction and replaces it with standing monotonicity: a standing preorder on raw stages whose quotient by standing-equivalence is a non-invertible partial order. The familiar arrows become admissible, plural, and physically interpretable only as effective projections or refinements of that quotient direction.

2 Core disambiguation: time, the standing arrow, and effective arrows

A common misreading treats every arrow word as the same object. The paper therefore fixes the following disambiguation before any theorem is used.

Object	AASC status	Primitive status
Physical time role T_D	Ordering of retained physical consequence in the realized interior.	Not primitive; licensed downstream of non-null realization, localization, comparison, transport, carrier, particle, gauge, mass, and vacuum roles.
Standing preorder $\preceq_S$	Raw-stage relation of standing retention, inheritance, conditioning, or transport.	Forced by irreversible standing fixation; not metric time.
Standing quotient order $\leq_S$	Partial order on standing-equivalence classes $\mathcal{E}_D/\equiv_S$.	The invariant arrow-bearing order.
Standing arrow $<_S$	Strict non-invertible direction on standing-equivalence classes.	Forced, but not a primitive physical substance or extra field.
Physical clock quotient $Q_{\text{Clk}}(D, U)$	Use-relative quotient class of certified same-scope clock-use behavior, treated in Paper II.	Not primitive; it depends on declared use, physical descent, readout or record burden, coherent continuation, and failure typing.
Record arrow	Asymmetric reuse of fixed witnesses.	Effective projection of standing fixation.
Causal arrow	Witnessed dependence across T_D .	Effective refinement, not primitive influence.
Thermodynamic arrow	Coarse-grained monotone over admitted state/coarse-graining structure.	Effective physical monotone, not the ground of standing.
Quantum measurement arrow	Record-fixation quotient over predictive dynamics.	Effective record structure; not primitive collapse unless richer dynamics are declared.
Psychological arrow	Agent-level memory/anticipation projection of record asymmetry.	Effective interior projection.

Object	AASC status	Primitive status
Cosmological arrow	Witness-availability, constraint-reachability, or boundary asymmetry inside a declared cosmological scope.	Effective or anchor-relative, not primitive admissibility source.

Thus T_D is the physical ordering role; the standing arrow is the invariant directionality condition; the physical clock quotient is a downstream use-relative carrier object; and effective arrows are domain-specific projections or refinements. A clock can represent, sample, or operationalize admitted temporal structure without grounding its direction. The paper does not identify time, clock use, causality, entropy, record-use, psychological flow, and cosmological asymmetry. It classifies their dependency order.

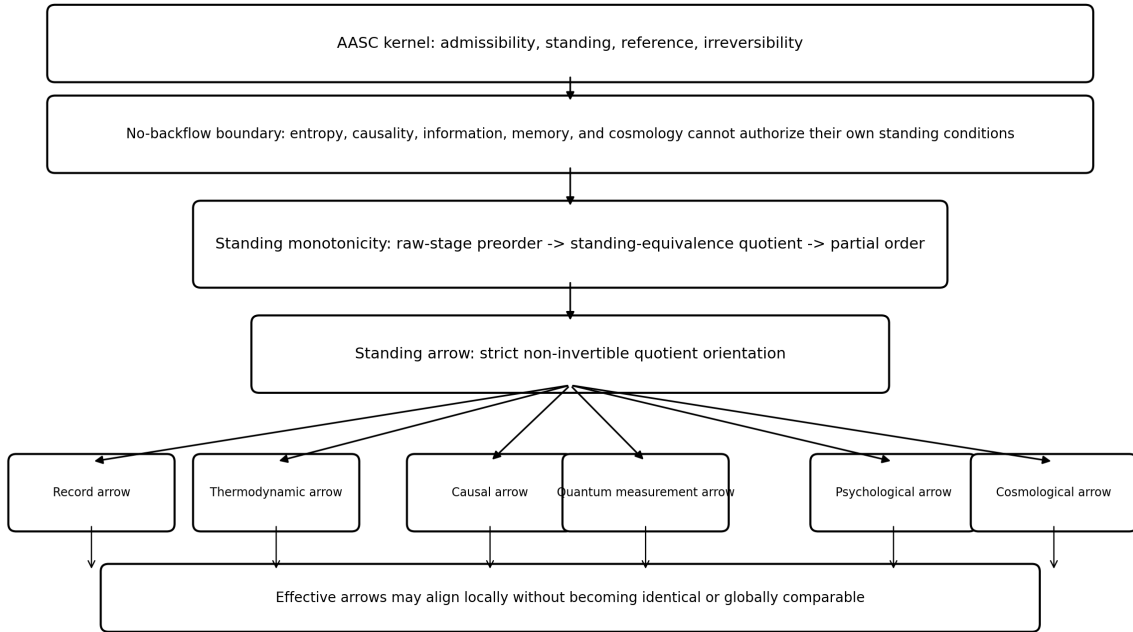


Figure 2: Standing monotonicity as invariant orientation, with familiar arrows as effective projections or refinements.

Figure 2 is schematic rather than generative. The upper layer records the AASC conditions that make standing-bearing directionality possible. The middle layer is the standing quotient order. The lower layer contains effective arrows that may align in a given physical regime without becoming identical or globally comparable.

3 The arrow-of-time problem under admissibility

The phrase “arrow of time” covers several distinct asymmetries. It can refer to the thermodynamic asymmetry between entropy-decreasing and entropy-increasing histories, the causal asymmetry between cause and effect, the evidential asymmetry between records and predictions, the psychological asymmetry between memory and anticipation, the quantum asymmetry between reversible predictive

dynamics and irreversible measurement records, or the cosmological asymmetry between one large-scale boundary direction and another. Ordinary discourse often treats these as one arrow because, in our physical regime, they frequently align. AASC does not begin with that alignment. It asks what must already be true for any such directionality claim to have standing.

The controlling distinction is between primitive and effective arrows. A primitive arrow is used as a foundation: it is supposed to ground the directionality of the regime without needing a prior standing-preserving structure. An effective arrow is admitted inside a domain whose standing, reference, redescription discipline, and irreversible continuation are already fixed. A primitive arrow is a ground. An effective arrow is a role, projection, monotone, refinement, or representation.

The question of this paper is therefore not “Which familiar arrow is most important?” The question is: which kind of directionality remains when no familiar arrow is allowed to authorize its own admissibility? The answer is standing monotonicity in quotient form. An irreversible standing-bearing commitment fixes a consequence that downstream admissible states may preserve, transform, condition on, or enrich, but may not erase as the same act without repair, scope change, or standing failure. On raw stages this gives a standing preorder; on standing-equivalence classes it gives a non-invertible partial order. The strict direction of that quotient order is the invariant arrow.

Definition 1 (Primitive arrow). *A primitive arrow is a directionality relation invoked as a ground of admissibility, standing, reference, record fixation, or irreversible continuation before those roles have been fixed.*

Definition 2 (Effective arrow). *An effective arrow is a directionality relation admitted inside a standing-bearing scope after the relevant reference, record, comparison, or continuation roles have standing. Effective arrows may be local, domain-relative, model-dependent, and plural.*

Definition 3 (Standing arrow). *The standing arrow is the strict non-invertible direction on the standing quotient induced by admissible irreversible fixation of standing-bearing consequence. It is not one more physical monotone among entropy, expansion, or causal order. It is the admissibility condition any effective arrow must respect.*

4 Dependency lock

The dependency order is not optional. The kernel is not inferred from time, entropy, causality, probability, information, records, clocks, or cosmology. The upstream kernel theorem establishes that any non-degenerate regime of admissibility-assessable, identity-preserving standing-bearing construction already instantiates admissibility, standing, reference, and irreversibility. This paper begins only after that necessity result is in force. Its local task is to prove the arrow-specific consequence for regimes that remain kernel-faithful: irreversible standing fixation induces a preorder on raw stages, a partial order on standing-equivalence classes, and a strict non-invertible standing arrow.

The paper imports five upstream AASC layers. Earlier source titles are cited verbatim where they preserve earlier naming conventions; the operative designation throughout this manuscript is AASC.

First, the meta-necessity layer states that non-degenerate irreversible commitment under unrestricted composition requires a standing-preserving admissibility invariant. If a decision can be erased by later recomposition without any tracked invariant, then the alleged irreversibility was undefined. If it cannot be erased, then a status-bearing distinction persists and must be governed.

Second, the kernel layer necessity-instantiates the constructional package

$$K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D).$$

Admissibility distinguishes licensed from unlicensed continuation. Standing gives an admitted act reusable status. Reference preserves the target of construction or physical content. Irreversibility blocks same-act repair from erasing failure or fixation. These roles are not premises selected for the arrow theorem; they are the forced governance structure of the non-degenerate target class.

Third, the AMetric/interface layer blocks primitive metricity, coordinates, scale, time parameters, probability measures, and representative choices from doing boundary work. Once admitted, such structures may become anchors, tensors, or skin. Before admission, they are hidden selectors.

Fourth, the realized-physical-interior layer supplies the physical role package

$$I_D = (\text{NonVacRec}_D, [B_D]_{\equiv B}, \text{Adj}_D, N_D, \text{LCD}_D, \text{Cmp}_D, \text{BoundTrans}_D, \Phi_D, \Pi_D, \\ G_D, M_D, \Omega_D, T_D, \text{Cau}_D).$$

The role order matters. Physical time is not primitive; it enters as ordered retained consequence after non-null realization, localization, comparison, bounded transport, fields, particles, gauge, mass, and vacuum roles have standing. Causality is not primitive; it enters as witnessed dependence across that order.

Fifth, the record-fixation layer states that empirical records become standing-bearing only through admissible record fixation. A mark is not evidence until it fixes what it records, survives lawful redescription, is reusable downstream, and preserves the difference between admissible and failed record use.

4.1 Irreversibility is not temporal flow

The irreversibility imported from the kernel is constructional non-erasure under same-scope redescription, not primitive temporal passage. The word “irreversible” is therefore the easiest place to misread the paper. In this paper, irreversibility is a standing condition, not primitive temporal flow. An act is irreversible in a fixed scope when its standing-bearing consequence cannot be erased, retroactively invalidated, or reclassified as the same act under lawful same-scope redescription. No clock, metric duration, flowing present, global foliation, or physical time parameter is required for that definition.

“Update” means a standing-indexed transition in an admissible construction or realized-continuation sequence. “Later” means position in a standing relation unless a downstream physical time representation has explicitly been licensed. “Retained consequence” means reusable standing-bearing status, not persistence through an independently flowing temporal medium. Physical time appears only at the role T_D , where the realized physical interior orders retained consequence. The pre- T_D argument therefore uses irreversibility as non-erasure under admissible redescription, not as a hidden time-arrow.

Proposition 4 (Non-temporal irreversibility). *Let D be a fixed admissible domain. An act is irreversible in D when its standing-bearing consequence cannot be erased, retroactively invalidated, or reclassified as the same act under lawful same-scope redescription. This condition requires admissibility, standing, reference, and no-repair; it does not require metric time, clock time, physical duration, or primitive temporal flow.*

Proof. Erasure, retroactive invalidation, and same-act repair are relations among standing statuses in a fixed scope. They can be evaluated by asking whether the target remains referenced, whether its admitted status remains reusable, and whether a failure or fixation has been laundered into validity without declared repair. None of those questions requires a metric parameter. If a later physical model represents the same structure by clock time, proper time, coordinate time, or a Hamiltonian parameter, that representation is downstream of the standing condition. Therefore irreversibility at this level is non-temporal non-erasure.

This establishes the non-circular dependency. The arrow theorem does not infer irreversibility from temporal flow. It begins with kernel irreversibility, which is forced by non-degenerate standing-bearing construction, and derives the quotient orientation that temporal and effective arrows must preserve.

A general schema governs all attempted primitive-arrow explanations. Any arrow capable of carrying standing-bearing directionality must have identifiable relata, stable reference for those relata, an admissible equivalence or comparison relation, reuse conditions, and failure conditions for misapplication. These requirements are already admissibility and standing requirements. An alleged arrow that lacks them is not yet a determinate arrow; an alleged arrow that has them cannot ground them without reversing the dependency order.

Theorem 5 (No effective-arrow backflow). *In a kernel-faithful regime, no downstream effective arrow may authorize the kernel conditions required to identify, compare, transport, or reuse that arrow as standing-bearing content.*

Proof. Any candidate arrow that carries directionality must first identify what is ordered, what counts as the same item under redescription, how comparison is made, how the ordering is reused, and what would count as misuse or failure. Those conditions are not supplied by the candidate arrow as a primitive; they are the standing and admissibility conditions that make the candidate arrow determinate.

The standard candidates instantiate this schema. Entropy requires identified states, a macro/micro or quotient ledger, comparison across stages, and usually a measure. Causality requires ordered relata and a witnessed dependence relation. Information requires encoding, reference stability, and an equivalence rule for what counts as the same message, state, or distinction under redescription. Cosmological asymmetry requires boundary, witness, or reachability identification. Psychological memory requires record fixation. Each candidate therefore presupposes standing, reference, and redescription stability in order to function as an arrow rather than as notation. If one of these candidates is used to ground standing itself, the dependency order is reversed. The result is hidden load, representative privilege, anchor smuggling, or rebranded admissibility. Therefore effective arrows may be admitted downstream, but they cannot backflow into the conditions that license them.

Backflow route	What it presupposes	Collapse mode
Entropy backflow	State space, macro/micro quotient, measure or ordering, comparison class.	Hidden standing or undeclared coarse-graining authority.
Information backflow	Encoding, reference stability, sameness of message/state, decoding criterion.	Rebranded admissibility or hidden reference.

Backflow route	What it presupposes	Collapse mode
Causality backflow	Ordered relata, dependence witness, no-pathless-influence rule.	Hidden temporal selector or hidden influence primitive.
Metric-time backflow	Comparison, clock/coordinate standing, representative discipline.	Metric selector before the AMetric boundary is crossed.
Cosmological backflow	Boundary reference, witness transportability, scope declaration.	Anchor smuggling or privileged initial-condition selector.
Memory backflow	Record fixation, evidence reuse, agent-level access.	Psychological projection promoted to primitive.

Remark 6 (Entropy backflow). *An entropy gradient can be a physical arrow only after the state space, macro/micro relation, coarse-graining, record ledger, and comparison class have standing. It cannot ground the standing conditions that make it an entropy gradient rather than a presentation-dependent bookkeeping choice.*

5 The realized-physics bridge to the arrow

The arrow paper is not a duplicate of the realized-physical-interior synthesis. The realized-physical-interior synthesis answers a role-architecture question: what roles must a physical regime instantiate for standing-bearing persistence to be non-null, localized, comparable, transportable, carrier-mediated, particle-classifiable, gauge-stable, deformation-responsive, vacuum-supported, temporally ordered, and causally dependent? The present paper answers a different question: once that architecture is available, what explains directionality and the familiar arrows?

The bridge matters because the arrow of time is often discussed as though it were a free-standing physical datum. AASC makes it dependent on the whole physical role package. A bare temporal label is not enough. A coordinate time without retained consequence is a numbering scheme. A metric interval without standing comparison is a representative. A causal cone without witnessed dependence is a geometric display. An entropy gradient without record-compatible coarse-graining is a scalar over an unlicensed quotient. A cosmological boundary without declared anchor status is an illicit selector.

A clock readout is likewise not the source of temporal direction. A clock is a certified carrier for ordering, conditioning, synchronizing, or comparing content inside a declared use. It may represent, sample, or operationalize physical time, but it cannot authorize the standing arrow or select its own admissibility conditions. Clock certification is therefore a downstream quotient problem rather than a primitive-arrow source.

The realized-physics ladder can be read as a sequence of preconditions for a physically meaningful arrow:

Realized-physics role	Arrow relevance
Non-vacuous realization	There must be something standing-bearing for directionality to concern.

Realized-physics role	Arrow relevance
Bounded localization	Records, perturbations, and restorations require loci of burden.
Local domains	Directional dependence and propagation require admissible neighborhoods and continuation domains.
Quotient-stable comparison	Metric, thermodynamic, and causal representatives require comparison that survives redescription.
Bounded perturbation transport	Causal and record relations cannot be pathless or global by default.
Fields as persistence carriers	Local persistence and records require carrier mediation rather than empty labels.
Particles as localized persistence classes	Object records require localized identity classes, not primitive point names.
Gauge as boundary-fixed equivalence	Presentation multiplicity must not become false physical multiplicity.
Mass/inertia as deformation response	Macroscopic irreversibility and source ledgers require typed response roles.
Vacuum as non-null baseline	The absence of particles cannot be identified with nullity or empty time.
Time as retained consequence	Physical time is licensed as irreversible continuation ordering.
Causality as witnessed dependence	Causal direction is a refinement, not the base ordering.

This bridge also controls explanatory priority. The paper does not say: entropy produces records, records produce time, time produces causality. It says: admissible standing-bearing persistence forces an irreversible order; records are fixed witnesses in that order; thermodynamic arrows are coarse-grained projections; causality is witnessed dependence; and cosmological arrows are large-scale reachability or boundary structures over the same admissible interior.

Proposition 7 (RPI import discipline). *The realized-physical-interior role ladder may be imported into the arrow paper only as role architecture, not as a primitive temporal premise.*

Proof. The RPI ladder derives temporal and causal roles after non-null realization, locality, comparison, transport, carrier, particle, gauge, mass, and vacuum roles. If the arrow paper used time or causality as primitive inputs, it would reverse that dependency order. Importing RPI as architecture preserves the dependency lock; using RPI to smuggle primitive time would violate it. Therefore RPI can supply the role basis for physical time and causality while the arrow-specific theorem must still prove why standing monotonicity, records, entropy, and cosmology are classified as they are.

The claim is not that every formal ordering requires the full realized-physical-interior stack. The claim is scope-specific: in a realized physical interior of the kind treated here, physical time as retained-consequence ordering is licensed only after the relevant physical persistence, comparison, transport, carrier, record-bearing, and baseline roles have standing. Abstract orderings, mathematical parameters, and reversible formal evolutions may exist in other registers; they become physical time only when they carry retained consequence inside an admitted physical interior.

6 Why the temporal and causal roles do not exhaust the arrow problem

The direct R11 theorem source gives the immediate temporal-causal role result: temporal parameter assignment before irreversible continuation ordering is inadmissible; reversible continuation without residue collapses fixation; record fixation is a boundary-fixed retained continuation witness; and causal dependence is induced when later standing-bearing content requires an earlier fixed witness. These are the correct local role theorems. They do not, by themselves, answer the broader arrow-of-time problem.

The broader problem includes at least six additional burdens.

First, it must classify entropy without either rejecting thermodynamics or making entropy primitive. Second, it must classify information without letting encoding replace standing. Third, it must distinguish causal order from causality and causality from metric cones. Fourth, it must explain why the past is fixed and the future open without importing metaphysical flow. Fifth, it must handle quantum predictive reversibility and measurement-record irreversibility in one account. Sixth, it must classify cosmological boundary conditions and large-scale arrows without turning initial conditions into hidden selectors.

Theorem 8 (TD/CauD sufficiency boundary). *The RPI temporal and causal roles are sufficient to license physical time and causal dependence as roles, but not sufficient to complete the explanatory classification of all arrows of time.*

Proof. T_D supplies ordered retained consequence and Cau_D supplies witnessed dependence across that order. These roles answer what time and causality are inside the realized physical interior. The arrow-of-time target asks which directionality relations can ground or explain entropy, records, memory, quantum measurement, cosmology, and effective asymmetry. Since these relations require additional quotient, witness, coarse-graining, boundary, and agent-level classifications, T_D and Cau_D fix necessary role content but do not exhaust the arrow-of-time classification.

7 Primitive-arrow route exhaustion: temporal direction

A primitive temporal direction says that there is a basic before/after orientation from which admissibility, records, causality, or physical irreversibility may be derived. AASC rejects that move, not because temporal language is useless, but because it cannot do pre-admissible work.

To say that one physical item is earlier than another in a load-bearing sense requires a stable target. The items must remain the same items under lawful redescription. Their order must not depend on a coordinate representative. The relation must be reusable by later reasoning. These are standing and reference requirements. If the relation is also directional, then reversal must fail in a way that preserves some consequence. That is irreversibility.

Theorem 9 (No primitive temporal direction). *Primitive temporal direction cannot function as a foundation of standing-bearing physical directionality in a kernel-faithful non-degenerate scope. Any such attempt either imports ordering or metric structure before standing, privileges a representative, or presupposes the irreversible fixation it was meant to ground.*

Proof. Assume a primitive temporal direction $<$ is introduced before admissibility, standing, reference, and irreversibility have been fixed. If $<$ is merely a label relation, it is skin and cannot ground physical directionality. If it is physically load-bearing, then it must identify ordered relata

across lawful redescription and permit downstream reuse; that requires reference and standing. If it distinguishes a direction rather than a symmetric relation, then reversal must fail in a retained way; that requires irreversibility. If a coordinate, clock, foliation, metric, or parameter is used to provide the relation, then representative or metric authority has been imported before the AMetric boundary permits it. Therefore the alleged primitive either does no work, performs work by hidden import, or presupposes the AASC roles it was supposed to ground.

Corollary 10 (Coordinate and proper time survive downstream). *Coordinate time, proper time, Hamiltonian parameters, clock readings, and foliations are admissible as downstream representatives once comparison, ordering, and record roles are licensed. They are not primitive standing generators.*

8 No primitive entropy

Entropy is indispensable inside thermodynamics and statistical mechanics. The AASC claim is narrower: entropy cannot be the primitive source of the arrow. Entropy requires an admitted state space, a macro/micro distinction or equivalent quotient, a coarse-graining rule, a comparison ledger, and an ordering over changes. None of these is pre-admissible.

Definition 11 (Admissible coarse-graining). *A coarse-graining is admissible in scope D when it quotients presentation or inaccessible distinctions without erasing standing-bearing tensor content required by the declared target. It may discard distinctions, but only those whose loss is licensed relative to the standing and witness ledger of the scope.*

Theorem 12 (Entropy cannot ground the arrow). *Entropy cannot function as a primitive ground of temporal directionality. Any entropy arrow presupposes standing-bearing state identification, admissible coarse-graining, reference-stable macrostates, and an ordering discipline.*

Proof. Let $S(x)$ be a proposed entropy assignment. If x is not a standing-bearing state, $S(x)$ is not a physical assignment. If the macrostate is not stable under lawful redescription, entropy changes may reflect presentation change rather than physical change. If coarse-graining is not admissible, loss of distinctions may erase tensor content rather than track effective macroscopic behavior. If there is no admitted ordering of stages, the statement that entropy increases has no directional carrier. Thus entropy can track an arrow only after the relevant standing, quotient, and continuation structures are fixed. It cannot ground those structures without circularity.

Remark 13 (Thermodynamic scope preservation). *The second law, thermodynamic modeling, entropy production, fluctuation theory, and statistical explanations remain fully operative inside their declared scopes. The excluded route is the foundational shortcut from an entropy scalar to primitive temporal direction.*

9 Probability, typicality, and measure status

Probability is often used to bridge reversible microphysics and irreversible thermodynamics. The AASC boundary is simple: probability is admissible as a post-standing modeling tool, not as primitive directionality. A probability measure requires a sample space, event identity, a sigma-algebra or equivalent event ledger, and a rule for comparing alternatives. None of these is pre-admissible.

Definition 14 (Post-admissible probability). *A probability assignment is post-admissible when its sample space, event identity, conditioning operations, and witness interpretation are fixed inside a standing-bearing scope.*

Theorem 15 (No primitive typicality). *Typicality cannot ground the arrow of time before standing is fixed. A typicality claim presupposes a measure or comparison class, and the measure presupposes event identity and admissible scope.*

Proof. A typicality statement says that some set of histories, states, or outcomes is large, generic, negligible, or expected relative to a measure or comparison class. If the histories or states are not referentially stable, typicality has no physical target. If the measure is not admitted, typicality is a hidden selector. If the conditioning data are not standing-bearing, the probability assignment cannot support record-use or prediction. Thus typicality can explain effective thermodynamic behavior only after the relevant probabilistic structure is admitted. It cannot ground admissibility itself.

This theorem is not anti-probability. It is a claim-status rule. Statistical mechanics may use measures, ensembles, typicality, fluctuations, and large-deviation results inside its declared scope. The AASC claim is that those tools enter downstream of standing, not upstream of it. A low-entropy macrostate may be atypical relative to a measure; but the macrostate, the measure, and the sense of atypicality must already be standing-bearing.

10 No primitive information

Information-theoretic accounts often appear more general than entropy-first accounts. They replace entropy increase with loss of information, growth of records, mutual information, algorithmic complexity, Bayesian evidence, distinguishability, compression, or encoding length. AASC treats all such quantities as post-admissible.

Theorem 16 (Information non-primitivity). *No informational quantity is admissible as an ontic primitive ground of temporal direction. Information presupposes encoding, reference stability, and redescription-invariant comparison.*

Proof. An informational quantity is always information of, about, between, or relative to something. That target must be referentially stable. The coding or representation scheme must be fixed or quotiented. The relevant equivalence class must preserve standing under redescription. Without these conditions, numerical information can change under recoding, relabeling, or choice of description without changing physical content. With these conditions, admissibility and standing have already been presupposed. Thus information may be an effective bookkeeping or witness measure, but it cannot ground the admissibility conditions required for an arrow.

Corollary 17 (No relabeling escape). *Replacing “information” by “state,” “encoding,” “distinguishability,” “complexity,” “compression,” “description length,” or “Bayesian evidence” does not recover primitivity unless the replacement term is promoted to ontic standing. Such promotion is hidden load unless independently admitted.*

11 No primitive causality

Causality is often used as the intuitive source of time’s direction: causes precede effects. But within AASC, causality itself is already a refined relation. Temporal order alone is not causality, and causality cannot be asserted without witnessed dependence.

Definition 18 (Witnessed dependence). *A later standing-bearing stage e_j witness-depends on an earlier standing-bearing stage e_i when some admitted witness of e_j requires, preserves, transports, or lawfully conditions on the fixed content of e_i under the declared scope.*

Theorem 19 (Causality cannot ground temporal direction). *Causal direction cannot be primitive. A causal relation either duplicates an ordering relation, imports hidden influence, or asserts dependence without a witness unless it is defined over an already admitted standing order.*

Proof. Suppose $e_i \rightsquigarrow e_j$ is primitive. If it means only that e_i is before e_j , it duplicates temporal order and cannot ground that order. If it means that e_i produces e_j , production must be witnessed or it is hidden influence. If it is witnessed, then the witness must have reference, standing, and redescription stability. If a metric light cone is used to ground causality, comparison and bounded transport have already been licensed. Therefore causal direction is an admissible refinement of standing quotient order, not its primitive source.

12 No primitive cosmological initial-condition arrow

The low-entropy universe, a special beginning, cosmological expansion, horizon asymmetry, or a privileged boundary condition may explain why many effective arrows align in our observed regime. Cosmological boundary data retain their physical importance as declared model anchors. The theorem excludes their use as primitive admissibility grounds.

Theorem 20 (No primitive initial-condition arrow). *A privileged initial state cannot ground temporal direction unless its privilege is itself standing-bearing under admissible reference, transport, and redescription. If so, it is an admitted boundary anchor inside a scope, not a primitive arrow.*

Proof. A state cannot be privileged unless it can be identified as the same boundary condition across descriptions and reused as a constraint. Identification requires reference. Reuse requires standing. Its role as a boundary rather than a display convention requires anchor status. If these are absent, the boundary condition is an arbitrary selector. If they are present, the boundary condition is downstream of admissibility. Therefore cosmological boundary data can be declared as anchors inside a model, but they cannot ground the conditions that make them admissible anchors.

Remark 21 (Past Hypothesis classification). *The Past Hypothesis may be a coherent cosmological anchor. It fails only if it is asked to be more than an anchor: namely, if it is used to supply the admissibility, standing, and reference conditions that allow it to be a boundary condition at all.*

Part II

Standing Monotonicity

13 Standing-bearing update and irreversible fixation

The positive construction begins with standing-bearing stages. These are not instants of primitive time. They are role configurations inside a fixed admissible scope.

Definition 22 (Standing-bearing stage). *A standing-bearing stage e in scope D is a role configuration whose content has admitted reference, reusable standing, and continuation relevance under D .*

Definition 23 (Standing fixation). *A standing-bearing content c is fixed at stage e_i when later admissible stages must either preserve, lawfully transform, explicitly repair, or explicitly abandon c by scope change. They may not erase the fact of fixation as the same act.*

Definition 24 (Retained consequence). *A retained consequence is fixed standing-bearing content that remains available for downstream admissible use, comparison, conditioning, record evaluation, or failure assessment.*

Lemma 25 (Fixation residue). *Every non-degenerate irreversible standing fixation leaves a residue: a standing-bearing difference between a scope in which the fixation occurred and a scope in which it did not.*

Proof. If no such difference exists, then the fixation makes no downstream difference and is degenerate. If the difference exists but can be erased as the same act without repair or scope declaration, the act was reversible in the relevant sense. Non-degenerate irreversibility therefore requires a retained standing-bearing residue.

14 Standing monotonicity

Standing monotonicity is the paper’s central theorem. It is stated in two stages for exactness. Raw standing-bearing stages carry a preorder. A genuine partial order appears only after quotienting by standing equivalence. This avoids the false requirement that distinct presentations or mutually reachable no-difference stages be antisymmetric before quotienting.

Definition 26 (Standing-bearing stage space). *Let $\mathcal{E}_D$ be the class of raw standing-bearing stages in a fixed admissible scope D . A stage may be a constructional commitment, a record-fixation status, a realized-continuation status, or a physical-history status, provided it carries reusable standing-bearing consequence in D .*

Definition 27 (Standing precedence). *For $e_i, e_j \in \mathcal{E}_D$, write*

$$e_i \preceq_S e_j$$

when standing fixed at e_i is retained, transported, presupposed, conditioned upon, or lawfully inherited in e_j .

Definition 28 (Standing equivalence). *Define*

$$e_i \equiv_S e_j \iff (e_i \preceq_S e_j) \wedge (e_j \preceq_S e_i).$$

Standing equivalence is mutual standing precedence. It is the equivalence relation used to form the quotient carrier on which antisymmetry is exact.

Lemma 29 (Mutual precedence absorbs non-residual difference). *If $e_i \equiv_S e_j$, then any alleged difference between e_i and e_j is either presentation skin, declared richer tensor load, scope change, hidden load, or a difference outside the fixed standing quotient. No same-scope strict standing residue remains between equivalent classes.*

Proof. Mutual standing precedence means that each stage retains, transports, presupposes, conditions upon, or lawfully inherits the standing of the other. If a strict same-scope residue remained between them, then one class would contain standing-bearing content not lawfully inherited by the other, contradicting mutual precedence. If an alleged difference changes no standing-bearing role, it is presentation skin. If it changes standing-bearing content while remaining undeclared, it is hidden

load. If it is declared, it is richer tensor load or a scope change. Therefore equivalence absorbs all non-residual differences inside the fixed quotient.

Definition 30 (Quotient standing order and strict standing arrow). *Let $[e]_S$ denote the $\equiv_S$ -class of e . Define*

$$[e_i]_S \leq_S [e_j]_S \iff e_i \preceq_S e_j,$$

and define the strict standing arrow by

$$[e_i]_S <_S [e_j]_S \iff ([e_i]_S \leq_S [e_j]_S) \wedge ([e_i]_S \neq [e_j]_S).$$

Definition 31 (Standing monotone). *A relation, quantity, or role is standing-monotone when its admissible downstream use cannot erase the standing-bearing consequence of its upstream fixation without repair, scope change, hidden selector, or standing violation.*

Theorem 32 (Standing Monotonicity Theorem, kernel-faithful quotient form). *Let D be a fixed kernel-faithful admissible scope with non-degenerate irreversible standing fixation. Every admissible irreversible commitment in D induces a standing preorder $\preceq_S$ on raw standing-bearing stages. Quotienting by standing equivalence $\equiv_S$ yields a partial order $\leq_S$ on $\mathcal{E}_D/\equiv_S$. The standing arrow is the strict non-invertible direction $<_S$ of this quotient order.*

Proof. The kernel is not produced by this construction. The proof begins after target instantiation and kernel faithfulness are in force and establishes their arrow-specific consequence. The nontrivial step is the quotient step. Kernel irreversibility forces retained residue and therefore supports composable standing precedence, but it does not license antisymmetry on raw stages. Distinct raw stages may be mutually precedence-equivalent because they differ only by presentation or by reversible no-difference continuation. The invariant arrow-bearing object is obtained only after those non-residual differences are absorbed by the quotient. Reflexivity holds because each standing-bearing stage retains its own fixed status under identity redescription. Transitivity holds because retained consequence composes: if e_i is retained, transported, presupposed, conditioned upon, or inherited in e_j , and e_j is so retained in e_k , then the standing-bearing consequence of e_i reaches e_k unless a declared repair, scope change, or failure intervenes. Thus $\preceq_S$ is a preorder on raw stages.

Raw-stage antisymmetry need not hold. Two stages may be mutually standing-precedent because they are different presentations of the same standing content or because reversible no-difference continuation connects them. The correct object is therefore the quotient by $\equiv_S$. On equivalence classes, antisymmetry is exact: if $[e_i]_S \leq_S [e_j]_S$ and $[e_j]_S \leq_S [e_i]_S$, then by definition $e_i \equiv_S e_j$, hence $[e_i]_S = [e_j]_S$. Reflexivity and transitivity descend from the preorder. Therefore $\leq_S$ is a partial order on standing-equivalence classes.

Finally, the strict direction $<_S$ is non-invertible. If $[e_i]_S <_S [e_j]_S$ and the strict reversal also held, the fixed residue from e_i to e_j would have to be erased, retroactively invalidated, reclassified as the same act, or neutralized by an undeclared selector. Those are exactly repair, standing violation, hidden load, or scope change. They are not same-scope admissible reversals. Therefore the standing arrow is the strict non-invertible direction of the quotient partial order.

Corollary 33 (The standing arrow). *The invariant direction of any admissible irreversible scope is not a primitive temporal flow and not a raw-stage antisymmetry. It is the strict quotient direction $<_S$ on standing-equivalence classes.*

15 Uniqueness of the invariant arrow

Effective arrows may be many. The invariant arrow is unique in the following precise sense: any admissible arrow must preserve standing and therefore factor through, refine, or locally project the standing quotient order.

The standing arrow is unique as an invariant orientation on the standing quotient, not as a global linear or total order. The uniqueness claim is not uniqueness of a global time parameter, physical clock, entropy functional, causal graph, or cosmological foliation. It is uniqueness of the invariant standing orientation that every admissible effective arrow must preserve, refine, or locally project in a kernel-faithful regime. Incomparable quotient classes are admissible unless a domain-specific projection supplies additional comparability.

Definition 34 (Admissible arrow). *An admissible arrow in scope D is a monotone ordering A_D such that: (i) it preserves standing under redescription; (ii) its load-bearing component is non-invertible in the role claimed; and (iii) its directionality does not depend on a hidden primitive source forbidden by the scope.*

Theorem 35 (Standing Arrow Uniqueness Theorem). *Every admissible arrow factors through, refines, or locally projects the standing quotient order $\leq_S$. No admissible arrow is independent of the standing quotient.*

Proof. Let A_D be an admissible arrow. If it does not preserve standing, it fails the first condition of admissibility. If it orders only distinctions that carry no standing-bearing consequence, its physical component is skin. If it orders standing-bearing distinctions, those distinctions correspond to retained, inherited, transported, or record-fixable residues among standing-equivalence classes. The arrow therefore factors through $\leq_S$, refines it by adding domain-specific structure such as causal witness dependence, or projects it through a coarse-grained quotient such as thermodynamic loss. Any alleged independent arrow must either violate standing, move scope, import a hidden selector, or collapse to presentation. Therefore no admissible arrow is independent of the standing quotient.

Remark 36 (Plurality without reduction to entropy). *The theorem does not say every effective arrow is numerically identical to the standing arrow. It says every effective arrow must be standing-compatible. Thermodynamic, causal, psychological, measurement-record, and cosmological arrows can locally align without being the same structure, and they can fail to globalize without threatening the invariant standing arrow.*

Part III

Records, Entropy, and Reversibility

16 Why the past is fixed

The common intuition that the past is fixed and the future is open receives a structural interpretation. The past is not fixed because a metaphysical flow has passed over it. It is fixed because it is the standing-fixed region of the continuation order.

Definition 37 (Past region of a scope). *For a stage e in scope D , the past region $P_D(e)$ is the class of standing-bearing stages p such that $p \preceq_S e$ and p contributes retained consequence, record*

availability, or admissible conditioning to e .

Theorem 38 (Past-fixation theorem). *The past of an admissible interior is the standing-fixed region of the continuation order.*

Proof. A stage counts as past relative to e only if it can be referred to, retained, or used in evaluating e . That is precisely standing-bearing fixation inherited by e . A merely labeled earlier coordinate that leaves no retained consequence is not past in the admissibility-bearing sense; it is a representative marker. Conversely, any standing-fixed retained consequence inherited by e functions as past relative to e , regardless of how it is represented by coordinates. Therefore pastness is standing fixation relative to the continuation order.

This gives a clean answer to the puzzle of fixed past without requiring primitive flow. The past is fixed because its records and consequences have standing. The fixity is not psychological; memory is downstream from record fixation. Nor is it merely thermodynamic; entropy may track many record processes, but record standing is not reducible to an entropy scalar.

17 Why the future is open

Future openness is likewise structural. It is not necessarily metaphysical indeterminism. It is the status of candidate continuation before standing fixation.

Definition 39 (Future candidate region). *For a stage e , the future candidate region $F_D(e)$ is the class of admissible or candidate continuations not yet standing-fixed relative to e .*

Theorem 40 (Future-openness theorem). *The future is open relative to an admissible interior because candidate continuations have not yet acquired standing-fixation relative to the current stage.*

Proof. A candidate continuation may be modeled, predicted, assigned probability, or represented by a deterministic equation. None of these statuses is the same as record fixation. Until the continuation's standing-bearing consequences are fixed and available for downstream reuse, it remains future-relative to the present stage. If all candidates were already fixed as records in the same admissible sense, the distinction between prediction and record would collapse. Since non-degenerate record use preserves that distinction, future openness follows as pre-fixation status.

Remark 41 (Determinism compatibility). *A deterministic dynamical law may assign a unique continuation from an initial data set. AASC still distinguishes predictive determination from standing-fixation. The future may be determined by a model without yet being record-fixed inside the admissible interior.*

18 Record asymmetry

Records are the load-bearing bridge between standing monotonicity and familiar temporal experience. A record is not merely stored information. It is a boundary-fixed retained continuation witness available for admissible downstream reuse.

Definition 42 (Record). *A record is a standing-bearing witness whose fixation survives lawful re-description and whose content can be reused in later admissible reasoning, comparison, measurement, or restoration.*

Theorem 43 (Record asymmetry). *Records can condition later admissible states only after standing-bearing fixation. Unfixed candidate continuations cannot perform the same record role.*

Proof. To condition a later state, a record must have fixed content and standing under redescription. A candidate future state lacks such fixation relative to the current stage. If it is treated as equally record-like, then prediction and record are identified, which erases the difference between standing-bearing witness and candidate model. If a future record is asserted to exist without fixation, it is hidden load. Therefore records are asymmetric: fixed records can condition later admissible states, while unfixed candidates cannot perform the same role.

Corollary 44 (Evidence points standing-backward). *Evidence is asymmetric because it is record-based. It refers to standing-fixed witnesses in the inherited order, not to unfixed candidate continuations.*

19 Memory, anticipation, and psychological time

Psychological time is an agent-level projection of record asymmetry. An agent remembers fixed records and anticipates candidate continuations. That contrast is powerful phenomenologically, but it is not primitive.

Proposition 45 (Psychological arrow as projection). *Memory and anticipation are downstream projections of record asymmetry and admissible prospective search. They do not ground temporal direction.*

Proof. Memory requires access to retained records. Anticipation requires modeling candidate continuations not yet fixed as records. Both presuppose the standing difference between fixed witness and unfixed candidate. Therefore psychological direction projects the record arrow into agent-level cognition; it cannot ground the record arrow.

The experience of flow may be a legitimate phenomenological or cognitive object. AASC classifies it as skin unless it is tied to standing-bearing updates, records, or action constraints. This is not a demotion of experience; it is a claim-status discipline. Experience may be real as an event in the physical interior while the phenomenological interpretation of flow is not allowed to ground the arrow.

20 Thermodynamic arrow as effective monotone

Thermodynamics supplies the most familiar physical arrow. AASC reclassifies it as effective. Entropy increase is a domain-relative monotone under admitted state spaces, coarse-graining, and dynamical assumptions. It is not the primitive origin of directionality.

Definition 46 (Thermodynamic effective arrow). *A thermodynamic effective arrow is a monotone tendency or ordering associated with standing-preserving coarse-graining, macroscopic stabilization, and irreversible loss of admissible distinctions in a declared thermodynamic scope.*

20.1 Entropy objects require admitted ledgers

The abstract claim that entropy is downstream becomes physics-facing when applied to standard entropy objects. The point is not that these objects are invalid. The point is that each requires a

ledger of admitted state identity, measure, quotient, or trace structure before it can carry directional content.

20.1.1 Boltzmann entropy

A Boltzmann entropy assignment has the schematic form

$$S_B = k_B \log \mu(\Gamma_M).$$

It requires a phase space or state space, a macroregion M , a measure μ on Γ_M , a macro/micro quotient, and a rule for comparing macrostates across continuation. In finite or discretized presentations this may be written schematically as $S_B = k_B \log |\Gamma_M|$; in the continuous case the measure is part of the admitted statistical ledger. Without those ledgers, $\mu(\Gamma_M)$ is not a standing-bearing physical quantity. With them, Boltzmann entropy is an excellent effective entropy object inside the declared statistical-mechanical scope.

20.1.2 Gibbs entropy

A Gibbs entropy assignment has the schematic form

$$S_G = -k_B \int \rho \log \rho d\Gamma.$$

It requires a probability density ρ , a measure $d\Gamma$, state identification, normalization, and an admitted ensemble or distribution interpretation. In continuous form the logarithm is meaningful only relative to a measure or reference-density convention; that convention is part of the admitted statistical ledger. Probability is not primitive at the AMetric boundary; it is a post-admissible structure over an identified sample or state space. Gibbs entropy therefore cannot ground standing, although it can carry standing once the statistical ledger is declared.

20.1.3 Von Neumann entropy

A von Neumann entropy assignment has the schematic form

$$S(\rho) = -\text{Tr}(\rho \log \rho).$$

It requires Hilbert-space state standing, a trace operation, a density operator, and usually a subsystem, algebra, partition, or measurement context. When used empirically it also requires record and calibration standing. It is therefore a quantum effective entropy object, not a primitive source of record fixation or temporal direction.

Theorem 47 (Entropy non-primitivity across standard forms). *Boltzmann, Gibbs, and von Neumann entropy each require admitted state, measure or reference-density, quotient, trace, or coarse-graining structure. Therefore none can ground the arrow of time as primitive. Each may serve as an effective thermodynamic or quantum-statistical monotone inside its admitted scope.*

Proof. S_B requires macroregions and a measure on the macroregion; S_G requires a probability density and measure; $S(\rho)$ requires Hilbert state standing and trace/subsystem discipline. Each structure presupposes reference stability and admissible comparison. A quantity that presupposes

those conditions cannot ground them. Once the ledgers are admitted, however, each entropy object can be used as a standing-bearing effective monotone in the corresponding scope.

Theorem 48 (Thermodynamic projection theorem). *Thermodynamic directionality is an effective projection of standing monotonicity through admissible coarse-graining. It is not a primitive arrow.*

Proof. Thermodynamic direction requires states, macrostates, coarse-graining, and comparison of macroscopic profiles. Each requires standing and reference. Irreversible macroscopic behavior tracks loss, compression, or inaccessibility of distinctions relative to the admitted coarse-graining. That loss can be monotone under the declared dynamics and can be represented by entropy or entropy-like quantities. But the monotone is defined over a standing-bearing quotient. Therefore it projects the standing arrow through a thermodynamic quotient rather than grounding it.

Remark 49 (Entropy/standing separation). *Standing is not scalar, additive, extensive, probabilistic, or macrostate-relative. Entropy often is. Thus entropy is not standing renamed. Entropy is a domain-specific measure over an admitted coarse-grained structure; standing is the admissibility condition for such measures to have physical content.*

21 Thermodynamic constraint ledger

The thermodynamic arrow has several distinguishable components that are often collapsed into one claim. AASC separates them to prevent overclaim.

Component	Required standing condition	AASC status
State space	Stable state identity under redescription	Anchor or tensor depending on scope.
Macrostate map	Admissible quotient from detailed states to macroscopic profiles	Coarse-graining role.
Entropy scalar	Invariant or convention-controlled assignment over macrostates	Effective measure, not primitive.
Equilibrium	Declared stability or extremal condition inside thermodynamic model	Model-specific tensor or anchor.
Dynamical mixing	Declared evolution or transport assumptions	Tensor load; not supplied by entropy alone.
Irreversibility	Non-invertibility of record standing or coarse-grained standing in the claimed role	Standing-monotone projection.
Fluctuations	Probability-bearing deviations inside an admitted measure	Post-admissible statistical content.

Proposition 50 (Entropy arrow evaluation). *A thermodynamic arrow claim is constraint-complete only when it declares its state space, coarse-graining, entropy functional or ordering, dynamical assumptions, record/witness relation, and scope of monotonicity.*

Proof. Without a state space, entropy has no target. Without coarse-graining, macroscopic loss is undefined. Without an entropy functional or ordering, no monotone is named. Without dynamics, no direction of change is specified. Without records or witnesses, the arrow cannot be empirically reused. Without scope, exceptions, fluctuations, boundary conditions, and open-system exchanges cannot be typed. Therefore all six declarations are required for constraint review completeness.

This is the place where AASC improves rather than weakens thermodynamic explanation. It prevents a scalar from carrying hidden work. A full thermodynamic account may still be excellent; it simply must declare which roles are doing which jobs.

22 Reversible dynamics and Loschmidt-style objections

A common objection is that microscopic equations may be reversible. If the underlying laws are reversible, where does the arrow come from? AASC answers by separating reversible representation from irreversible standing fixation.

Theorem 51 (Formal reversal is not standing reversal). *A reversible dynamical representation does not eliminate the standing arrow unless it also supplies an admissible reversal of record fixation, retained consequence, and standing-bearing witness reuse within the same scope.*

Proof. Let U_t be a reversible representative dynamics. Formal reversibility means that for a state representation x , $U_{-t}U_tx = x$ under the model. Standing reversal is stronger. It requires undoing the record quotient status, retained consequence, and downstream admissible uses of the stage as the same act. A formal inverse trajectory does not by itself erase those records. If the inverse is declared to erase them, it must alter the record object, change the scope, or perform repair. If it does not erase them, it is a reversible presentation over a non-reversed standing ledger. Therefore reversible dynamics does not erase standing monotonicity.

Loschmidt-style objections therefore locate a real distinction but not a refutation. Reversible microdynamics can coexist with a standing arrow because equations and records perform different roles. The equation may permit formal inversion; the record ledger determines which fixed witnesses are reusable as evidence inside the admitted interior.

Part IV

Quantum, Causality, and Relativity

23 Quantum predictive reversibility and record irreversibility

Quantum theory intensifies the distinction. The predictive interior may be reversible and unitary, while measurement records are fixed in a quotient of record use. AASC does not need primitive collapse to obtain an arrow, nor does unitary reversibility erase records.

Definition 52 (Measurement-record arrow). *The measurement-record arrow is the effective directionality induced when quantum-sensitive predictive structure produces or supports standing-bearing records available for downstream reuse.*

Theorem 53 (Measurement-record arrow). *Quantum measurement directionality enters through standing-bearing record fixation, not through primitive collapse, primitive consciousness, or primitive branching.*

Proof. A measurement outcome functions as an outcome only if it is a record: it fixes what is recorded, survives admissible redescription, and can be reused. A collapse postulate may be declared as richer-scope dynamics; hidden variables may be declared as richer-scope state data; branching may

be interpreted through record-equivalence classes; operational descriptions may restrict themselves to record quotients. In each case, same-scope standing-bearing outcome content is carried by record fixation. Without record fixation, there is no evidence-bearing outcome. Therefore the measurement arrow is a record arrow, not a primitive ontological selector.

Interpretive route	AASC classification
Collapse dynamics	Richer-scope dynamics if explicitly declared; hidden selector if used without declaration.
Hidden variables	Richer-scope state data if they carry standing-bearing witness content; otherwise skin.
Everett branching	Admissible only through record-equivalence and branch-use profiles; branch labels alone are skin.
Operationalism	Record quotient without additional ontology claim.
Consciousness collapse	Hidden selector unless formally declared as tensor load with witness conditions.

24 Decoherence, environmental records, and classicality

Decoherence is an important physical mechanism and should not be conflated with the AASC record quotient. The relationship is:

$$\text{decoherence} \neq \text{record standing},$$

but

$$\text{decoherence} + \text{record-fixation quotient} \Rightarrow \text{admissible classical record use}.$$

Decoherence can explain why interference between certain branches, pointer profiles, or environmental correlations becomes effectively suppressed. It can supply stability for record candidates. Decoherence supplies the physical stabilization mechanism. AASC fixes the distinct standing question: when has a physical mark, pointer profile, environmental correlation, or measurement trace become reusable evidence under lawful redescription?

The record-standing boundary is operationally fixed by the following criterion.

Criterion	Standing role
Readout fixation	The mark, pointer profile, environmental imprint, or trace fixes determinate content.
Reference and calibration	The record remains about a stable target under the declared measurement and readout relation.
Redescription invariance	Lawful changes of presentation preserve the record's standing-bearing content.
Downstream evidential reuse	Later admissible reasoning may condition on the record as a fixed witness.

Criterion	Standing role
Quotient preservation	Equivalent record presentations preserve the same standing-bearing use profile.
Failure typing	Misread, erased, uncalibrated, merely decohered, or non-reusable candidates are typed as failed or incomplete record candidates rather than silently promoted to records.

Decoherence may support stability and redundancy within this criterion; it is not identical to the criterion. Record standing begins only when the record-use quotient and its failure boundary are fixed.

Theorem 54 (Decoherence/standing separation). *Decoherence may supply effective physical stability for record candidates, but admissible record use requires record-fixation quotienting. Therefore decoherence can support the measurement arrow but cannot replace the standing conditions that make a record usable as evidence.*

Proof. A decoherence model supplies dynamical suppression of interference relative to an environment, pointer basis, partition, or coarse-grained profile. Those structures may stabilize record candidates, but a stabilized candidate is not yet a standing-bearing record. To be a record, it must fix what it records, survive lawful redescription, be reusable in later reasoning, and preserve the difference between admissible and failed record use. Those are quotient and standing conditions, not merely dynamical suppression conditions. Therefore decoherence supports record standing at the mechanism layer, while quotient fixation supplies the standing condition.

This distinction also prevents a false conflict. If a decoherence program declares its environment, pointer basis, interaction Hamiltonian, coarse-graining, and record ledger, it can provide the physical mechanism by which certain record candidates become robust. AASC then classifies which of those candidates have standing. If the declarations are absent, decoherence cannot by itself authorize a record; if they are present, decoherence is a downstream physical support for the record arrow.

25 Bell witnesses and nonlocality

Bell-type correlations are often read as challenging ordinary causal direction. AASC classifies them through joint standing rather than primitive signal arrows.

Proposition 55 (Bell records do not supply primitive temporal direction). *Bell-type correlations are tensor-level compatibility on joint standing classes. They do not supply a same-scope signal arrow and do not ground temporal direction.*

Proof. A Bell witness set contains invariant joint constraints not reducible to independently sufficient local standing data in the same operational scope. If the joint constraints are retyped as same-scope signals, the theorem object changes or hidden influence is imported. If they are treated as mere local labels, tensor-level joint compatibility is erased. The admissible reading is that local records sample a joint standing class. That classification is compatible with standing monotonicity but does not ground it.

26 Temporal order is not causality

Temporal order and causality must remain distinct. If every earlier stage caused every later stage, causality would collapse into succession. If causality did not require order or witness, it would become hidden influence.

Definition 56 (Causal dependence). *For stages e_i, e_j in scope D , write*

$$e_i \rightsquigarrow_D e_j$$

when e_j witness-depends on e_i across the standing quotient order.

Theorem 57 (Causal refinement theorem). *Causality is an admissible refinement of standing monotonicity by witnessed dependence.*

Proof. If $e_i \rightsquigarrow_D e_j$, then the later content of e_j requires fixed content from e_i through an admitted witness. Thus $e_i \prec_S e_j$. But not every $e_i \prec_S e_j$ is causal; some earlier fixed content may be merely background, coincident, or independent. The witness-dependence condition refines the standing quotient order by selecting dependence-bearing pairs. Therefore causality is a refinement, not the base arrow.

Corollary 58 (Metric cones are downstream). *Relativistic causal cones, propagator support, and finite-signal structures are mature presentations of comparison, transport, temporal ordering, and causal dependence. They cannot ground those roles at the AMetric boundary.*

27 Causal models, interventions, and counterfactuals

Causal models often define direction through interventions or counterfactual dependence. AASC permits such models as effective causal frameworks, provided their intervention variables, background conditions, admissible manipulations, and witness ledgers are declared. An intervention is not primitive magic; it is a scoped operation with standing-bearing consequences.

Definition 59 (Admissible intervention). *An intervention is admissible in scope D when it changes a declared tensor or anchor according to a lawful operation while preserving the record, comparison, and witness conditions required to evaluate downstream dependence.*

Proposition 60 (Intervention status). *Intervention-based causality refines standing quotient order; it does not ground it.*

Proof. An intervention must specify what is changed, what is held fixed, which outcomes are measured, and how counterfactual alternatives are compared. These require reference, standing, and scope. The causal direction inferred from intervention is therefore a witnessed-dependence relation over an admitted experimental or model structure. It refines the standing quotient order but cannot ground it.

This classification also handles counterfactuals. A counterfactual is not a record of a future or alternate world. It is a candidate comparison inside a model scope. Its usefulness depends on which anchors and tensors are held fixed. Without that ledger, counterfactual causality becomes a hidden selector.

28 Retrocausality, delayed choice, and causal loops

AASC does not forbid every formal model with two-boundary constraints, retrocausal language, or closed curves. It forbids same-scope retroactive standing repair.

Theorem 61 (No same-scope retroactive standing repair). *No causal model may retroactively alter standing-fixed record content within the same admissible scope.*

Proof. A standing-fixed record is available for downstream reuse because its content is stable under lawful redescription. If a later act changes that content as the same act, then the earlier record was not fixed, or the later act is a repair, or the scope has changed. Same-scope retroactive alteration therefore violates record standing. Models with two-boundary conditions, delayed-choice experimental contexts, or closed timelike curves may be declared as richer-scope structures, but then they must supply consistency, boundary, and record certificates.

Route	AASC status
Formal two-boundary model	Possible richer-scope anchor if both boundaries and record status are declared.
Retrocausal influence changing a fixed record	Standing violation or repair.
Block-model correlation	Representation of an ordered structure, not primitive cause.
Delayed-choice experiment	Record-context dependence, not retroactive repair of fixed standing.
Closed timelike curve	Requires richer-scope consistency certificate and no-record-contradiction ledger.

29 Relativity, simultaneity, and metric time

Relativity is not an obstacle to standing monotonicity. It is a warning against primitive global simultaneity. Coordinate time, proper time, foliation choice, and clock readings are representatives inside admitted metric/comparison structure.

Proposition 62 (Metric time as representative). *Coordinate time, proper time, foliations, and Hamiltonian parameters are downstream representatives of admitted ordering and comparison structure.*

Proof. A coordinate labels; a clock compares; a proper time integrates along a worldline inside a metric projection; a foliation slices a representative spacetime. Each presupposes admitted comparison and standing. Therefore these structures can represent temporal order but cannot supply it pre-admissibly.

Relativity of simultaneity fits naturally. Simultaneity is foliation- or frame-relative skin unless tied to invariant causal or record structure. Standing order need not be a global simultaneity order. It is a partial order of retained consequence and witnessed dependence.

30 The block universe objection

AASC does not refute block representation. A global block may encode the entire ordered structure. The issue is not metaphysical becoming. The issue is role asymmetry within the represented interior:

record-use, evidential access, causal dependence, and standing-fixation are not symmetric roles merely because the global representation contains all stages.

Theorem 63 (Block representation does not collapse standing role). *A representation that contains all stages of an ordered physical structure does not make all stages equivalent with respect to record-use, evidential access, causal dependence, or standing-fixation. It collapses the standing arrow only if it also collapses those role distinctions into skin.*

Proof. A block representation may assign coordinates or geometric placement to all events in a model. But record-use is not coordinate inclusion; it is the standing-bearing ability of a fixed witness to be reused in admissible reasoning. Evidential access is not mere existence in a representation; it is availability under the interior’s record and transport conditions. Causal dependence is not universal containment; it is witnessed dependence across the temporal role. If the block preserves these distinctions, it preserves the standing arrow internally. If it denies them, it changes the target from a record-bearing physical interior to a skin-only global display. Therefore block representation is compatible with AASC but does not trivialize standing monotonicity.

Block-universe claim	AASC response
All events are represented in one mathematical object.	Compatible. AASC does not require primitive flow.
There is no global moving present.	Compatible. The standing arrow is not a moving-present thesis.
Simultaneity is foliation-relative.	Compatible. The standing arrow is not global simultaneity.
Past/future are representationally perspectival.	Partly compatible; representation perspective does not erase record-role asymmetry.
All represented events have the same standing role.	Denied unless record-use, evidence, causal dependence, and fixation are all collapsed to skin.

The compressed reply is: “all events are represented” does not imply “all events play the same standing role relative to a record-bearing interior.”

Part V

Cosmology and Plural Effective Arrows

31 Cosmological arrow as witness-availability asymmetry

A cosmological arrow is not primitive temporal direction at large scale. It is an effective asymmetry in witness availability, constraint reachability, boundary data, horizon access, or record-supporting structure inside a declared cosmological scope.

Definition 64 (Cosmological effective arrow). *A cosmological effective arrow is a large-scale monotone or orientation in admissible witness availability, constraint reachability, boundary relation, record density, or horizon access across the standing quotient order.*

Cosmological asymmetry	AASC classification
Expansion	Metric/cosmological representation; not primitive arrow by itself.
Low-entropy boundary	Declared model anchor or hidden selector if undeclared.
CMB record structure	Standing-bearing witness layer inside the observed cosmological interior.
Structure formation	Effective constraint/reachability process over admitted dynamics and records.
Horizon asymmetry	Witness-access boundary requiring a horizon/record ledger.
Black-hole evaporation	Richer physical arrow requiring horizon, radiation, record, and leakage ledgers.
Heat death	Thermodynamic/cosmological effective limit, not primitive temporal source.
Cosmological time parameter	Representative after metric, comparison, and ordering roles are licensed.
Inflationary initial data	Model anchor or boundary condition, not primitive admissibility source.

Classification 65 (Cosmological effective-arrow classification). *A cosmological arrow is admissible only as an effective asymmetry in witness availability, constraint reachability, boundary data, horizon-access structure, or large-scale record organization inside a declared cosmological scope. If it is used to ground admissibility itself, it becomes anchor smuggling or hidden selector import.*

Argument. To state a cosmological asymmetry, the model must identify the large-scale states or histories being compared, the boundary or horizon conditions relevant to the comparison, the witnesses or records through which the asymmetry is known, and the metric or dynamical structure used for reachability. These are admitted ledgers. If the asymmetry is introduced after those ledgers, it is an effective cosmological arrow. If it is used before those ledgers to ground admissibility, it privileges a boundary, representative, or initial condition without standing. That is anchor smuggling or hidden selector import.

The Past Hypothesis is therefore classified, not dismissed. As declared boundary data in a cosmological/statistical model it is an anchor. As a primitive ground of standing it backflows. AASC permits the first use and rejects the second.

32 Vacuum, zero-point offsets, and cosmological-arrow discipline

Vacuum is central to cosmological arrow discussions because one can confuse no-particle baseline, zero-point scalar offsets, dark-energy ledgers, and cosmological expansion. AASC separates these. The realized physical interior treats vacuum as a non-null no-particle persistence baseline. The zero-point-energy discipline treats absolute homogeneous offsets as non-standing unless they descend to the relevant quotient or appear as matched boundary/effective ledger content.

Proposition 66 (Vacuum is not empty time). *The vacuum role does not supply primitive temporal direction. It supplies the non-null no-particle baseline over which fields, transport, records, and cosmological ledgers may have standing.*

Proof. If vacuum were total nullity, it could not support field-carrier structure, boundary traces, perturbation transport, or record formation. If it were a primitive temporal medium, it would import standing before the role ladder licenses it. The admissible vacuum role is the minimal concrete persistence baseline. It allows arrows to be represented in no-particle or homogeneous regimes without turning vacuum into time-flow.

Proposition 67 (Zero-point offset status). *A regulator-dependent absolute zero-point scalar does not supply a cosmological arrow. Relational vacuum responses, phase differences, and matched homogeneous cosmological ledgers may have standing inside declared scopes.*

Proof. An absolute offset changed by admissible renormalization or convention-preserving redescription while preserving witnesses is skin in the restricted homogeneous sector. Relational responses and matched effective terms can descend to the quotient. Since an arrow requires standing-bearing directionality, a non-standing scalar offset cannot ground cosmological direction. Declared relational or boundary quantities may contribute to effective cosmological models.

33 Horizons, reconstruction, and witness access

Horizons are relevant to arrow questions because they structure what can be witnessed, reconstructed, or transported. AASC treats horizon claims as reconstruction and witness-access claims, not primitive temporal direction claims.

Definition 68 (Witness-access asymmetry). *A witness-access asymmetry is a difference in the availability, reconstructability, or transportability of admissible witnesses across the standing quotient order.*

Proposition 69 (Horizon-arrow classification). *A horizon or horizon-like structure may contribute to an effective cosmological arrow by constraining witness access or reconstruction, but it cannot serve as a primitive arrow without a standing certificate.*

Proof. A horizon claim identifies what can or cannot be observed, reconstructed, transported, or causally contacted. These notions presuppose records, witnesses, transport, and comparison. If a horizon structure is declared in a physical model, it may alter effective witness availability and thereby support a cosmological arrow. If it is used to ground directionality before witness standing is fixed, it is hidden boundary load.

34 Plural cosmological arrows

Cosmology contains many arrows that may align in our regime but need not be identical in principle: expansion arrow, radiation/cooling arrow, structure-formation arrow, horizon/witness arrow, thermodynamic arrow, and record arrow. AASC allows plural cosmological arrows precisely because it refuses to collapse all directionality into one primitive scalar.

Proposition 70 (No mandatory global alignment). *Effective cosmological arrows need not globally align. Their alignment is a physical fact or model condition, not an admissibility primitive.*

Proof. Each cosmological arrow has a different role basis: scale-factor behavior, temperature evolution, gravitational clustering, causal or horizon access, entropy/coarse-graining, or record availability. Since these bases are distinct effective structures over standing quotient order, no

AASC theorem forces their global identity. Local or epochal alignment may be explained by physical dynamics, but it is not a primitive requirement.

35 Definition of effective arrows

The preceding sections define several arrow types. We now collect the general criterion.

Definition 71 (Effective arrow). *An effective arrow in scope D is a relation or monotone A_D satisfying all of the following:*

1. *It preserves standing under lawful redescription.*
2. *It is non-invertible in the role it claims to track.*
3. *It induces, depends on, or is compatible with record asymmetry.*
4. *It does not function as a primitive ground of its own admissibility.*
5. *Its scope, anchors, tensors, and skin are declared.*

Arrow	Basis	Global alignment required?	Primitive?
Standing arrow	Irreversible standing fixation	Invariant	Forced, not primitive source
Record arrow	Boundary-fixed retained witnesses	Local standing alignment	No
Thermo-dynamic arrow	Coarse-grained loss or entropy-like monotone	No	No
Causal arrow	Witnessed dependence	No	No
Psychological arrow	Memory/anticipation asymmetry	No	No
Quantum measurement arrow	Record-fixation quotient	No	No
Cosmological arrow	Witness reachability, boundary structure, expansion/cooling contexts	No	No

36 Plural effective arrows theorem

The standing arrow is unique as an invariant orientation on the standing quotient, not as a global linear or total order. The uniqueness claim is not uniqueness of a global time parameter, physical clock, entropy functional, causal graph, or cosmological foliation. It is uniqueness of the invariant standing orientation that every admissible effective arrow must preserve, refine, or locally project in a kernel-faithful regime. Incomparable quotient classes are admissible unless a domain-specific projection supplies additional comparability.

Theorem 72 (Plural Effective Arrows Theorem). *The standing arrow is unique and invariant. Thermodynamic, causal, record, psychological, quantum-measurement, and cosmological arrows are effective projections or refinements and need not globally align.*

The standing arrow is unique in this theorem as quotient orientation, not as totalization. Effective arrows may linearize, branch, coarse-grain, or leave incomparable parts of the standing quotient according to their own domain ledgers.

Proof. The standing arrow is unique by the Standing Arrow Uniqueness Theorem: every admissible arrow must preserve standing and hence factor through or refine standing quotient order. Thermodynamic arrows add admissible coarse-graining; causal arrows add witness-dependence; record arrows add retained witness reuse; psychological arrows add agent-level memory and prospective search; quantum measurement arrows add record quotienting over predictive dynamics; cosmological arrows add large-scale witness and reachability structure. Since the added structures differ, their effective arrows need not be identical or globally aligned. Since each is admissible only by standing preservation, none is independent of the standing arrow.

Part VI

Concrete Physical Illustrations and Effective Strength

37 How effective arrows acquire practical strength

Standing monotonicity is not itself a thermodynamic mechanism, a cosmological dynamics, or a decoherence model. It is the invariant orientation that such mechanisms must respect once their records, state spaces, comparison relations, and boundary conditions are admitted. Effective arrows acquire practical strength when several domain-specific structures reinforce the same standing orientation.

In our observed regime, the reinforcement is unusually robust. Macroscopic coarse-grainings lose distinctions in ways stable under ordinary intervention. Environmental decoherence proliferates record candidates into many surrounding degrees of freedom. Cosmological expansion and horizon structure organize witness availability and constraint reachability on large scales. Biological and cognitive systems inherit fixed records as memory and treat unfixed continuations as anticipation. These mechanisms are not identical, and AASC does not require their global alignment in all possible domains. It explains why, when they do align locally, their alignment is usable rather than primitive: each mechanism has a standing ledger that prevents its arrow from authorizing its own prerequisites.

The following examples are compact classification audits rather than sector-complete model derivations. They show how the exhaustion and role typing operate in familiar physical settings.

38 Case study I: entropy and Loschmidt reversal

Consider a classical gas described by a microstate in phase space and by a macrostate selected by a coarse-graining. A Boltzmann entropy assignment uses a macroregion Γ_M and a measure μ , schematically

$$S_B = k_B \log \mu(\Gamma_M).$$

This expression already assumes a state space, a macro/micro quotient, a measure convention, and a rule for reusing the macro-description across stages. If those ledgers are admitted, entropy may define an effective thermodynamic arrow. If they are not admitted, the symbol S_B has no standing-bearing directionality to ground.

Loschmidt reversal presses the standard worry. If the microscopic equations permit a reversed trajectory, why does entropy increase in the ordinary direction? AASC separates two issues. Formal reversibility of a trajectory is a statement about a dynamical representation. Standing reversal would require admissible erasure or reversal of record fixation, macrostate reuse, and retained consequence in the same scope. A reversed formal solution does not by itself reverse the record ledger that made the original entropy assignment usable as evidence.

Thus AASC does not solve statistical mechanics by denying reversibility. It classifies the reversible formal trajectory and the irreversible record/coarse-graining ledger as different roles. The thermodynamic arrow is strong in ordinary macroscopic practice because the admitted coarse-graining, environmental records, preparation constraints, and intervention limits reinforce the same standing orientation. That strength is physical and explanatory, but it is not primitive.

39 Case study II: decoherence, pointer records, and measurement direction

Consider a system S , apparatus A , and environment E in a measurement-like interaction. Decoherence theory explains how environmental entanglement suppresses interference between robust pointer-like alternatives and distributes correlations into many environmental degrees of freedom. In quantum Darwinism language, selected states can leave multiple redundant imprints in the environment.

AASC accepts this as a physical support mechanism for record candidates. It does not identify decoherence itself with record standing. A decohered correlation becomes a record only when it fixes what it records, survives lawful redescription, can be reused in later evidential or deliberative contexts, and has a quotient status that distinguishes tensor content from presentation skin. In compressed form:

$$\text{decoherence} \neq \text{record standing},$$

while

$$\text{decoherence} + \text{record-fixation quotient} \Rightarrow \text{admissible classical record use}.$$

This distinction also clarifies delayed-choice and related experiments. A later measurement context may determine which record quotient is realized for a given experimental arrangement. It does not retroactively repair a fixed record inside the same scope. If a proposed account adds collapse dynamics, hidden variables, many-world branch ontology, or additional state data, AASC treats that as a declared richer scope. If no such addition is declared, the standing-bearing content is the record quotient.

The measurement arrow is therefore not a primitive quantum collapse arrow. It is the asymmetry between predictive dynamics and standing-bearing record fixation. Decoherence helps explain why record candidates become stable and widely accessible; quotient standing explains when they become admissible records.

40 Case study III: the Past Hypothesis as cosmological anchor

The Past Hypothesis says, in broad form, that the universe began in a very special low-entropy macrocondition. In statistical-mechanical explanations it can play a powerful role: it supplies an asymmetric boundary condition that helps explain why ordinary macroscopic entropy gradients point in the observed direction.

AASC does not reject that use. It classifies it. If the Past Hypothesis is declared as boundary data in a cosmological/statistical model, it is an anchor of that model's scope. The relevant questions then concern what state space, measure, macrocondition, and transport ledger the model declares. The boundary may be physically important, but it is not pre-admissible. Its own privilege must be reference-fixed and transported under admissible description.

If, by contrast, the Past Hypothesis is used as the primitive ground of directionality itself, it backflows. To call one boundary **past**, 'low entropy,' or "special" already requires comparison, coarse-graining, recordability, and standing of the cosmological description. The condition can be an anchor inside a model; it cannot be the source of the admissibility discipline that makes it an anchor.

This classification changes the explanatory burden. A cosmological arrow may be strong because expansion, horizon access, radiation history, structure formation, and record density all reinforce a common witness-availability gradient. But the origin of that boundary structure remains a downstream cosmological problem. AASC supplies the scope discipline for stating that problem without turning boundary selection into a hidden temporal primitive.

Part VII

Main Theorem and Route Exhaustion

41 Arrow architecture

For constraint-formalism purposes, define the arrow architecture of an admissible realized physical scope D as

$$\mathcal{A}_D = (S_D, R_D, C_D, H_D, Q_D, K_D, P_D),$$

where:

- S_D is the standing quotient order $\leq_S$ on $\mathcal{E}_D / \equiv_S$;
- R_D is the record-fixation quotient and record-reuse structure;
- C_D is the causal refinement by witnessed dependence;
- H_D is the thermodynamic/coarse-graining effective monotone family;
- Q_D is the quantum measurement-record layer when the scope includes quantum predictive structure;
- K_D is the cosmological reachability, witness-density, and boundary-anchor layer when the scope is cosmological;

- P_D is the psychological or agent-level projection layer when agents are included.

This tuple is not a new ontology. It is a constraint device. It records which role carries which directionality claim and prevents the common error of letting one layer silently do another layer's work.

42 Main AASC Arrow-of-Time Theorem

Theorem 73 (AASC Arrow-of-Time Theorem, exhaustion form). *Let D be a fixed kernel-faithful realized physical interior with standing-bearing persistence, irreversible continuation, record-fixable witnesses, lawful redescription, and no same-act repair. Every same-scope attempt to ground temporal direction by primitive time, metric time, entropy, causality, probability, information, memory, collapse, clock choice, or cosmological boundary authority resolves as backflow, hidden load, declared anchor or tensor import, effective projection or refinement, representative privilege, repair, scope change, or failure. The surviving architecture satisfies:*

1. raw standing-bearing stages carry a standing preorder $\preceq_S$;
2. standing-equivalence classes carry a quotient partial order $\leq_S$;
3. the standing arrow is the strict non-invertible direction $<_S$ of that quotient order;
4. primitive temporal direction, primitive entropy, primitive information, primitive causality, primitive probability, and primitive initial-condition arrows are inadmissible as foundations of standing-bearing directionality;
5. physical time T_D is represented as ordered retained consequence inside the realized physical interior;
6. causality Cau_D is witnessed dependence across T_D ;
7. records induce asymmetric reuse because fixed witnesses can condition downstream admissible states;
8. thermodynamic direction is an effective coarse-graining monotone; Boltzmann, Gibbs, and von Neumann entropy each require admitted ledgers and cannot function as primitive grounds;
9. quantum measurement directionality enters through record-fixation quotienting, with decoherence as possible physical support but not as record standing itself;
10. information measures are post-admissible encoding artifacts;
11. cosmological arrows are effective reachability, witness-density, boundary-anchor, horizon-access, or large-scale record asymmetries;
12. plural effective arrows are admissible and need not globally align;
13. no effective arrow may backflow into the admissibility conditions that license it;
14. physical clock objects, when realized, are downstream use-relative readout or conditioning carriers; their certification is not required to ground the standing arrow, and their plurality does not create primitive temporal direction.

Proof. The theorem does not derive or regenerate the kernel. Its hypothesis fixes a non-degenerate realized physical target that remains kernel-faithful, and each clause is a downstream consequence

within that comparison class. Items (1)-(3) follow from the quotient-form Standing Monotonicity Theorem. Item (4) follows from the primitive-arrow boundary eliminations and no-backflow theorem. Item (5) follows from the imported realized-physical-interior role of time as ordered retained consequence. Item (6) follows from the imported causal role as witnessed dependence across order. Item (7) follows from the Record Asymmetry Theorem. Item (8) follows from the Thermodynamic Projection Theorem and the standard entropy-ledger analysis. Item (9) follows from the Measurement-Record Arrow Theorem and the decoherence/standing separation theorem. Item (10) follows from Information Non-Primitivity. Item (11) follows from the Cosmological effective-arrow classification and the cosmological taxonomy. Item (12) follows from the Plural Effective Arrows Theorem. Item (13) follows from no effective-arrow backflow. Item (14) follows from the distinction between invariant directionality and downstream clock-use carriers: any clock with physical force must already satisfy admissibility, standing, reference, readout or record, continuation, and failure conditions, and therefore cannot ground the arrow that licenses its use. Since all clauses preserve the same fixed-scope standing discipline under lawful redescription, the theorem package is coherent.

43 Exhaustion as positive identification

The eliminative form of the theorem is its positive content. AASC does not select standing monotonicity from a menu of competing arrow mechanisms. It fixes the comparison class, denies downstream structures boundary-authorizing status, and exhausts every same-scope primitive route. What survives that closure is the invariant arrow-bearing structure.

Standing monotonicity is therefore not a preferred explanation placed beside entropy, causality, records, clocks, or cosmology. Those structures remain available as effective projections, refinements, representatives, or declared anchors after admission. The quotient direction is forced because no governance-non-equivalent same-scope primitive source remains. In a constraint formalism, survivor closure is positive identification.

A request for an additional primitive mechanism is a substantive challenge only if it preserves the fixed target and kernel faithfulness while escaping the route-exhaustion normal forms. Otherwise the request adds downstream physics, changes scope, imports hidden authority, or renames work already performed by the standing quotient.

44 Operational minimality and deletion matrix

Theorem 74 (Arrow-role minimality). *No major role in the arrow architecture can be deleted while preserving a non-degenerate account of directionality.*

Proof. The proof is by deletion. Delete admissibility and there is no licensed/unlicensed distinction. Delete standing and there is no reusable fixed status. Delete reference and the ordered relata are unstable. Delete irreversibility and no fixation residue remains. Delete records and the past/evidence asymmetry collapses. Delete coarse-graining discipline and entropy becomes presentation-dependent. Delete witnessed dependence and causality collapses to succession or hidden influence. Delete AMetric discipline and metric time becomes a hidden boundary selector. Delete no-backflow and effective arrows become illicit primitives. Therefore the architecture is operationally minimal.

Deleted role	Failure
Admissibility	No licensed/unlicensed continuation distinction.
Standing	No reusable fixed status and no retained consequence.
Reference	No stable target for before/after, record, or cause.
Irreversibility	No non-invertible fixation residue.
Record quotient	No evidence, memory, or past-fixation asymmetry.
Coarse-graining discipline	Entropy becomes presentation-dependent or tensor-erasing.
Witness dependence	Causality collapses to bare succession or hidden influence.
AMetric boundary	Metric time, coordinates, or scale can act as hidden selectors.
No-backflow rule	Effective arrows authorize their own prerequisites.

45 Closure against primitive-arrow challenges

Theorem 75 (Arrow architecture closure). *The AASC arrow architecture is closed under same-scope primitive-arrow challenges. Every same-scope primitive alternative is either a downstream effective arrow, skin, hidden load, anchor motion, tensor import, scope change, repair, representative privilege, or collapse.*

Proof. Let X be a proposed primitive arrow. If X preserves no standing-bearing content, it is skin. If it preserves content by an undeclared mechanism, it is hidden load. If it changes boundary conditions, it is anchor motion. If it adds new dynamics or state variables, it is tensor import or richer scope. If it changes lawful continuation, it changes scope. If it erases fixed records, it is repair. If it relies on coordinates, labels, entropy displays, or branch names as physical authority, it is representative privilege. If it supplies all required roles without changing scope, it is a presentation of the standing arrow or an effective arrow. No same-scope primitive alternative remains.

46 Proof-register note

The local results are fixed-domain constraint theorems, not lower derivations of the AASC kernel. Their hypotheses include target instantiation and kernel faithfulness; their conclusions state what is forced once the necessity-instantiated kernel, no-repair condition, AMetric boundary, and lawful-redescription discipline remain in force. Their proof mode is exhaustion: every same-scope alternative must either preserve the declared roles or terminate in a normal-form branch.

The paper uses several proof registers. A **theorem** states a same-scope necessity, quotient, impossibility, or closure result. A **proposition** states a dependency or classification consequence. An **argument** or **classification** identifies the role of a standard physics structure without claiming a sector-complete dynamical derivation. A **theorem boundary** marks work assigned to a richer or downstream target. This separation is part of the claim-status discipline: entropy, decoherence, cosmology, and quantum records retain their physical roles while remaining unable to backflow into the standing conditions that license them.

The appropriate refutation burden follows the theorem class. A critic must deny target instantiation, break a load-bearing upstream result, show an invalid specialization, or exhibit a faithful same-regime survivor. Demanding a preferred mechanism, familiar primitive, or new representative does not engage the theorem unless it supplies that survivor.

Numbering note. Definitions, propositions, theorems, lemmas, corollaries, remarks, and classifications

share a single formal-statement counter. The numbering therefore records the manuscript-wide order of formal statements rather than a separate sequence for each statement type.

47 Theorem ladder digest

For reference, the full theorem ladder can be compressed as follows.

1. **No Primitive Temporal Direction.** A primitive before/after relation cannot ground standing because it already requires stable ordered relata.
2. **No Entropy-as-Ground.** Entropy requires admitted state, macrostate, coarse-graining, and order.
3. **No Primitive Typicality.** Probability and typicality require admitted event identity and measure scope.
4. **Information Non-Primitivity.** Information requires encoding, reference, and redescription stability.
5. **No Primitive Causality.** Causality requires witnessed dependence across order.
6. **No Primitive Cosmological Boundary.** Boundary privilege requires anchor standing.
7. **Standing Fixation.** Non-degenerate irreversible commitment leaves a retained residue.
8. **Standing Monotonicity.** Retained residue induces a standing preorder on raw stages and a quotient partial order on standing-equivalence classes.
9. **Standing Arrow Uniqueness.** Every admissible arrow factors through or refines the standing quotient order.
10. **Past Fixation.** The past is the inherited standing-fixed region.
11. **Future Openness.** The future is candidate continuation not yet record-fixed.
12. **Record Asymmetry.** Fixed witnesses condition downstream use; unfixed candidates do not.
13. **Thermodynamic Projection.** Entropy arrows are effective coarse-graining monotones.
14. **Reversibility Non-Erasure.** Reversible equations do not erase record fixation.
15. **Measurement-Record Arrow.** Quantum measurement direction is record quotienting over predictive dynamics.
16. **Causal Refinement.** Causality is witnessed dependence over the standing quotient order.
17. **Cosmological Effective Arrow.** Cosmological arrows are witness/reachability/boundary structures.
18. **Plural Effective Arrows.** Effective arrows are plural and need not globally align.
19. **Architecture Closure.** Same-scope primitive-arrow rivals collapse to normal forms.

This digest functions as proof-role control. Any additional arrow claim must map to one of these theorem roles or declare a new scope.

48 Primitive-source route exhaustion

The final negative theorem is arrow-specific. Same-scope primitive-arrow proposals do not survive as foundations. They either become effective arrows inside standing monotonicity, change the theorem object, import hidden load, move an anchor, privilege a representative, perform repair, or collapse to skin.

Primitive-arrow route	Collapse mode	Admissible survivor
Primitive temporal flow	Hidden order selector or metaphysical surplus.	T_D as retained-consequence ordering role.
Primitive metric time	Metric backflow before comparison is licensed.	Coordinate/proper/foliation time as representative.
Primitive entropy	Entropy backflow through undeclared state/coarse-graining ledgers.	Thermodynamic effective monotone.
Primitive probability	Sample-space or typicality selector before admitted measure.	Statistical tool inside declared state space.
Primitive information	Encoding/reference backflow.	Post-admissible information measure.
Primitive causality	Hidden influence or hidden temporal ordering.	Witnessed-dependence refinement of T_D .
Primitive memory	Record backflow through agent access.	Psychological projection of record asymmetry.
Primitive low-entropy initial condition	Anchor smuggling.	Declared cosmological boundary condition.
Primitive collapse	Undeclared dynamics or branch/record selector.	Richer-scope dynamics or record quotient.
Primitive block	Representation absolutism.	Global encoding compatible with standing-role asymmetry.
Primitive complexity growth	Measure/backflow or computational selector.	Effective computational/statistical monotone.

Branch completeness. The route list is complete because every proposed primitive-arrow source must either preserve the fixed kernel-faithful comparison class or alter it. If it alters that class, it is theorem-object change, scope change, anchor motion, or declared tensor import. If it preserves the class, it either changes standing-bearing directionality or it does not. A change with standing-bearing force must operate through fixation, records, comparison, reference, a causal witness, an entropy or probability ledger, an information encoding, a boundary anchor, a metric representative, or a clock readout; each locus is already typed by the preceding architecture as a downstream effective or declared role. A putative source that changes none of those loci changes no admissible arrow fact and is skin or bookkeeping. No third same-scope primitive-arrow locus remains.

Theorem 76 (Same-scope primitive-arrow exhaustion). *Every same-scope primitive-arrow proposal either becomes an effective arrow inside standing monotonicity, changes the theorem object, imports hidden load, moves an anchor, performs repair, privileges a representative, or collapses to skin.*

Proof. A primitive-arrow proposal must carry directional physical content. If it preserves standing and has no independent load, it is an effective arrow or skin. If it changes the domain, boundary, dynamics, state space, probability measure, record selector, or cosmological condition, it moves an anchor or imports tensor load. If it chooses a direction without declaring the relevant ledger, it is a hidden selector or representative privilege. If it reverses fixed standing as the same act, it performs repair. The table enumerates the arrow-specific routes. No same-scope primitive source remains.

49 Counterexample burden

The paper is falsifiable at two distinct levels. An *internal* counterexample must preserve the same non-degenerate, kernel-faithful realized physical target and then violate the arrow-specific conclusion. An *external* rival may declare a different primitive package, but it first bears the target-instantiation question: if it supports the same admissibility-assessable, identity-preserving standing-bearing construction, the kernel roles are already in force by necessity; if it does not, it addresses a different target rather than furnishing a same-regime counterexample.

A responsive internal counterexample must therefore exhibit a regime D satisfying all of the following conditions.

1. D supports determinate standing-bearing physical content and preserves admissibility, standing, reference, and irreversibility.
2. D preserves lawful redescription, no same-act repair, and the declared realized-physical comparison class.
3. D admits non-degenerate irreversible fixation and record reuse.
4. D defines a genuine, load-bearing directionality relation.
5. Within that same scope, either the standing-precedence relation fails to form the asserted preorder and quotient partial order, or an admissible arrow survives that neither factors through, refines, nor locally projects the standing quotient.
6. The survivor is obtained without hidden selector work, primitive-arrow backflow, record repair, representative privilege, undeclared tensor import, anchor motion, or scope change.

Non-responsive examples include a reversible equation by itself, an entropy gradient without a state/coarse-graining ledger, a causal graph without witnessed dependence, a block representation without record-role analysis, a memory appeal without record fixation, an initial condition without boundary-status declaration, a collapse postulate without scope declaration, or an information measure without encoding/reference stability. Such examples may define legitimate structures in their own scopes; they do not preserve the comparison class required to refute the present theorem.

Theorem 77 (Counterexample normal form). *A responsive internal counterexample must preserve the non-degenerate, kernel-faithful realized physical target while either defeating the standing preorder/quotient-order result or exhibiting a same-scope admissible arrow independent of the standing quotient. Any proposal that requires primitive-arrow backflow, hidden selector work, repair, representative privilege, undeclared load, anchor motion, or scope change is not such a counterexample.*

Proof. Physical directionality that is load-bearing must preserve identifiable relata, lawful redescription, reusable consequence, and a stable distinction between fixation and repair. Those conditions

place the proposal inside the kernel-faithful comparison class. Within that class, a counterexample must attack either the order construction or the uniqueness/factorization conclusion. If it instead supplies direction by entropy, information, causality, probability, memory, metric time, clock choice, or cosmological boundary privilege acting as upstream authority, it violates no-backflow. If it changes records, dynamics, state space, boundary data, or the invariant bundle, it changes scope. If it supplies no standing-bearing difference, it is skin or degenerate. The listed routes therefore exhaust responsive internal challenges.

Part VIII

Objections, Comparison, and Conclusion

Each objection below has standing only if it supplies a same-scope primitive-arrow survivor that preserves kernel faithfulness while escaping standing monotonicity and the effective-arrow normal forms. Otherwise the proposal is a declared anchor, effective arrow, richer scope, hidden-load route, representative, repair, or skin.

50 Objection: update smuggles time

The strongest linguistic objection is that the paper uses words such as update, later, retained, consequence, and continuation. The reply is strict: these are standing-indexed terms, not primitive metric-time terms.

An update is a transformation in a governed construction or realized-continuation sequence. A later stage is later only relative to a standing quotient order. Retention means downstream availability of fixed content, not movement through an absolute temporal medium. Continuation is role preservation under admissible transformation, not passage of time as a substance.

The charge of smuggling would succeed only if the proofs required metric duration, global simultaneity, primitive flow, or an unlicensed before/after relation. They do not. They require non-invertible standing fixation. Physical time is introduced only after that order is available. A responsive defeat must therefore exhibit a same-scope temporal primitive that carries standing-bearing force without presupposing, importing, or reproducing the kernel.

51 Objection: this denies the second law

The second law remains intact as a physical theorem inside its declared scope. What is excluded is entropy's status as primitive ground of directionality. A thermodynamic system may have a robust entropy-increase theorem under declared assumptions. That theorem remains a physical theorem inside its scope. AASC imposes the prior constraint check: the state space, measure, macro/micro quotient, coarse-graining, and record ledger must already have standing.

Thus the second law survives as a downstream physical theorem family. What fails is the foundational move from "entropy increases" to "entropy grounds temporal direction before admissibility." A responsive counterexample must make entropy authorize its own state, measure, coarse-graining, comparison, and record ledgers while preserving the same kernel-faithful target.

52 Objection: causality already explains the arrow

Causality explains many physical asymmetries once causal dependence has standing. But causal dependence cannot be primitive in the relevant sense. To call one event a cause and another an effect requires more than order. It requires a witness of dependence. Without that witness, causality is either bare succession or hidden influence. With that witness, record standing and ordered retained consequence are already present.

Therefore causality is a powerful effective arrow, not the invariant arrow itself. A primitive-causal formalism may be declared as a richer theorem object; to refute the same-scope result it must preserve kernel faithfulness while grounding direction without ordered relata, reference, or witnessed dependence.

53 Objection: quantum mechanics is reversible

Quantum predictive dynamics can be reversible. The measurement-record layer is not identical to the reversible predictive layer. A unitary model may track state evolution; a record quotient tracks which outcomes become standing-bearing evidence. If a formal inverse exists but records remain fixed, the standing arrow remains. If records are undone, the model must declare repair, erasure dynamics, or scope change.

This is why reversible quantum dynamics and measurement directionality coexist without primitive collapse. A responsive counterexample must reverse standing-fixed records as the same act without repair, hidden erasure dynamics, or scope change.

54 Objection: cosmology explains the arrow through initial conditions

Cosmology may explain why many effective arrows align. A low-entropy boundary, expansion, radiation cooling, and structure formation may be physically central. But a boundary condition must itself have declared standing. Otherwise it is a hidden selector. AASC therefore classifies cosmological boundary data as anchors in a model, not as pre-admissible sources of directionality.

Cosmological arrows therefore survive as effective reachability, witness-density, and boundary-conditioned structures. A responsive counterexample must make a cosmological boundary privileged before its reference, anchor status, witness transport, and scope have standing.

55 Objection: information is more fundamental than time

Information cannot be more fundamental than the standing conditions that make information a determinate quantity. Encoding requires a scheme. A scheme requires reference to what is encoded. Comparison requires stability under redescription. A bit string without an admitted target is not physical information; it is a mark. Therefore information measures are post-admissible artifacts or tools, not primitive grounds. A responsive counterexample must fix encoding, reference, decoding, and sameness under redescription without already performing the governance work of the kernel.

56 Objection: standing monotonicity is just entropy renamed

Standing monotonicity is not entropy renamed. Entropy is a scalar or ordering defined over a state/coarse-graining structure. Standing is the reusable status of fixed content under admissible continuation. Entropy may be additive, extensive, probabilistic, coarse-grained, or ensemble-relative. Standing is none of these. It is a constraint on admissible reuse and redescription.

The relation is hierarchical: entropy arrows presuppose standing; standing does not presuppose entropy. A responsive counterexample must exhibit a scalar, additive, probabilistic, or macrostate-relative standing object that preserves the same target without changing the standing quotient.

57 Objection: standing monotonicity is too abstract to explain physics

AASC's abstraction is load-bearing: it classifies which roles must be present before physical structures are interpretable. The realized-physical-interior arc connects standing to locality, metric comparison, bounded transport, fields, particles, gauge, mass, vacuum, time, causality, and records. The present paper extends that classification to arrow-of-time questions.

Abstraction is a virtue here because the standard arrows are heterogeneous. Entropy, causality, quantum measurement, memory, and cosmology do not share one physical mechanism. They share a standing constraint: each must be admissible, record-compatible, and non-invertible in its role. A responsive objection must identify standing-relevant physical content outside this constraint architecture while preserving the same kernel-faithful regime.

58 Objection: clocks define time

A clock can define a scale, coordinate, synchronization convention, proper-time reading, relational parameter, or record-ordering practice inside a declared physical scope. It does not thereby ground the standing conditions that make those readings identifiable, reusable, and direction-sensitive. A cyclic clock may have local clock standing; a noisy clock may remain usable within a performance envelope; and several inequivalent clocks may coexist for different uses. None of those facts creates or selects the standing arrow.

Conversely, standing monotonicity does not choose a preferred clock. It fixes invariant orientation wherever standing comparison is available. The companion clock paper asks which candidate carriers descend to the physical quotient, satisfy a declared readout or record burden, remain coherent through continuation, and fail in typed ways. That is a clock-certification problem, not the source of temporal direction.

The compressed distinction is: clock standing is not temporal direction; temporal direction is not clock selection. A responsive counterexample must show a certified clock generating the standing arrow without importing metric authority, record backflow, selector privilege, or a richer theorem object.

59 Comparison with standard accounts

AASC operates at the constraint layer over standard accounts. It role-types their state spaces, measures, records, causal ledgers, boundary conditions, and representational structures, and it

prevents any effective component from being promoted beyond its declared scope. The native physical work of statistical mechanics, decoherence theory, causal modeling, and cosmology remains intact inside those scopes.

Standard accounts therefore survive in their native registers as declared anchors, standing-bearing tensor content, representatives, mechanisms, or effective arrows. AASC’s narrower and stronger claim is that none of those declared structures may backflow into the kernel conditions that make them identifiable, reusable, and standing-bearing. The exhaustion theorem role-types their load; it does not erase their sectoral physical work.

Account	Claimed ground	AASC assessment
Boltzmann and statistical mechanics	Entropy/probability	Valid effective arrow under admitted state, measure, and coarse-graining; not primitive.
Past Hypothesis	Low-entropy boundary	Declared cosmological anchor; not pre-admissible ground.
Causal asymmetry	Cause/effect	Admissible refinement by witnessed dependence; not primitive.
Records and information	Memory and stored information	Record fixation yes; information primitive no.
Quantum collapse	Measurement irreversibility	Richer-scope dynamics if declared; otherwise record quotient.
Decoherence	Environment-induced classicality	Effective record/coarse-graining layer; not final record standing by itself.
Block universe	Global spacetime representation	Compatible representation; does not erase internal standing relations.
Psychological arrow	Subjective flow or memory	Agent-level projection of record asymmetry.
Cosmological expansion	Large-scale asymmetry	Effective reachability/witness asymmetry, not primitive temporal flow.
Complexity growth	Algorithmic or structural complexity	Effective measure after encoding and comparison are admitted.
Quantum Darwinism	Redundant environmental records	Physical support for record candidates; standing still requires record quotienting.
Fluctuation theorems	Nonequilibrium work and entropy-production statistics	Effective statistical constraints inside admitted ensemble and measurement scopes.
Relational or thermal time	State-dependent temporal flow or modular evolution	Representative/effective time structure after the relevant algebraic and state ledgers are fixed.

Boltzmann, Gibbs, Loschmidt-style reversibility, and Past Hypothesis accounts remain indispensable comparators for statistical-mechanical arrow explanations. Their native derivational programs remain intact. AASC classifies the ledgers they require—state identification, measure, coarse-graining, boundary status, and record-bearing reuse—and blocks those ledgers from serving as their own pre-admissible authorization.

Price, Albert, Reichenbach, Eddington, and related time-asymmetry accounts correctly show that arrow explanations must control where asymmetry enters. AASC sharpens that control into a no-backflow condition: an asymmetry may be a declared anchor, an effective monotone, or a witnessed dependence relation, but it may not authorize the admissibility conditions that make it intelligible.

Decoherence accounts associated with Zeh, Zurek, and Joos-type environmental analysis are likewise retained as physical mechanisms for effective classicality and record stability. Quantum Darwinism sharpens this point by emphasizing redundant environmental records. AASC adds a distinct quotient question: when does a physically stabilized or redundantly imprinted mark become a standing-bearing record available for evidential reuse?

Past-Hypothesis and typicality programs, including recent work on the logical status of the Past Hypothesis, remain natural comparators because they ask how irreversible macrodynamics is obtained from time-reversal-compatible microdynamics. That derivational problem remains a downstream statistical-mechanical task. AASC classifies the Past Hypothesis, typicality measure, and macrostate partition as anchors or ledgers whose standing must be declared and cannot be generated by their later explanatory success.

Fluctuation theorems and large-deviation approaches are likewise compatible with the present account. They quantify constrained deviations and nonequilibrium work statistics inside an admitted experimental and probabilistic scope. They do not remove the need for record fixation, ensemble standing, and measurement ledgers.

Relational and thermal-time approaches are also compatible at the representative level. They may identify useful temporal parameters or flows from algebraic, state-dependent, or thermodynamic structure. AASC asks the upstream question of when those structures have standing and why their induced temporal roles cannot function as primitive sources of admissibility.

60 What AASC proves by exhaustion

AASC's distinctive contribution is its theorem class: fixed-domain route exhaustion, impossibility closure, and same-regime survivor analysis. The following six controls are not methodological preferences; they are the closure structure forced by the declared target.

First, it supplies a no-backflow criterion. Effective arrows cannot authorize their own prerequisites.

Second, it separates invariant standing direction from effective physical arrows. This permits plural arrows without forcing premature reduction to entropy, causality, or cosmology.

Third, it gives a route-exhaustion matrix for primitive-arrow accounts. This makes objections type-checkable.

Fourth, it explains past/future asymmetry through record fixation and standing-bearing candidate status.

Fifth, it separates reversible equation form from irreversible standing fixation.

Sixth, it places quantum measurement, thermodynamics, and cosmology in one constraint architecture without making them identical.

61 Theorem boundary and exact scope

The paper establishes a full-strength fixed-domain directionality result. Its theorem object is constraint closure rather than sector-complete numerical dynamics. In every non-degenerate realized physical regime that remains kernel-faithful, standing monotonicity is the forced invariant orientation and the familiar arrows are downstream effective structures. The following boundaries assign constructive and numerical work to the scopes in which it belongs; they do not weaken the quotient conclusion.

First, the origin of the low-entropy boundary of our universe remains a cosmological problem. The present theorem classifies low-entropy boundary claims as declared anchors inside cosmological/statistical scopes rather than primitive sources of admissibility.

Second, entropy-production rates, fluctuation probabilities, relaxation times, and large-deviation estimates remain the work of Boltzmann, Gibbs, von Neumann, fluctuation-theorem, and related statistical tools inside admitted scopes.

Third, decoherence and quantum Darwinism supply physical mechanisms for stability and environmental proliferation of record candidates. AASC fixes the quotient-standing condition under which those candidates become admissible records.

Fourth, global alignment of effective arrows is a physical or model-dependent result, not a condition of the standing quotient. The standing arrow is an invariant partial orientation; domain-specific projections may add comparability or fail to align in particular regimes.

Fifth, horizon structure, black-hole evaporation, cosmological reconstruction, and quantum-gravity records require richer scope declarations and additional witnesses. The constraint result fixes the admissibility architecture within which those programs are stated.

Sixth, physical-clock certification and the multiple-choice problem for internal time form the distinct downstream target of Paper II, *The Physical Clock as a Quotient*. A coordinate, matter field, subsystem, semiclassical parameter, modular flow, or record sequence acquires clock standing only through a declared use, physical descent, readout or record burden, coherent continuation, and failure typing.

These boundaries define the adjacent research program: mechanism-specific applications of the same constraint discipline to statistical mechanics, quantum records, cosmological boundaries, horizon witness access, and the conditions under which local effective arrows acquire approximate global alignment.

62 Conclusion

The arrow of time is not eliminated. It is fixed by dependency. Non-degenerate standing-bearing construction instantiates the AASC kernel by necessity; a realized physical regime that remains kernel-faithful then forces standing monotonicity as its invariant orientation. Primitive temporal direction cannot ground that structure. Entropy, probability, information, causality, memory, quantum collapse, and cosmological initial conditions cannot serve as primitive foundations of standing-bearing directionality without backflow. Each requires standing, reference, redescription stability, record fixation, or irreversible continuation before it can function as an arrow.

What is forced instead is standing monotonicity. An admissible irreversible commitment fixes a

standing-bearing consequence. Later admissible states may inherit, condition on, transform, or enrich that consequence, but may not erase its fixation as the same act without repair, scope change, or standing violation. That non-invertible quotient partial order is the invariant arrow.

The familiar arrows survive. The past is the standing-fixed region of the continuation order. The future is candidate continuation not yet record-fixed. Records point standing-backward because fixed witnesses condition downstream use. Entropy tracks effective loss under admitted coarse-graining. Causality refines the standing quotient order by witnessed dependence. Quantum measurement direction arises through record quotienting. Psychological time projects record asymmetry into memory and anticipation. Cosmological arrows express witness availability, constraint reachability, and declared boundary anchors.

In compressed form:

$$\begin{aligned} &\text{Irreversible standing fixation} \Rightarrow \text{standing monotonicity} \\ &\Rightarrow \{\text{record, thermodynamic, causal, quantum,} \\ &\quad \text{psychological, cosmological effective arrows}\}. \end{aligned}$$

AASC operates at the constraint layer over physics. It identifies what physical theories must preserve for their arrows to be arrows. The invariant survivor is not time-flow, entropy, information, causality, or cosmology. It is standing monotonicity.

Within the AASC time non-primitivity sequence, this is Paper I. It closes the directionality question, not the clock-choice question. Paper II, *The Physical Clock as a Quotient*, treats clock carriers as use-relative quotient objects and classifies clock identity, plurality, failure, and selection without restoring primitive time. Together the papers give a two-sided result: no primitive temporal direction and no primitive preferred clock.

The endpoint is survivor closure, not mechanism preference. After the same-scope primitive-arrow routes are exhausted, no invariant orientation remains outside the standing quotient. Subsequent applications should be judged by mechanism-specific clarity: whether a statistical-mechanical model makes its coarse-graining and typicality anchors explicit; whether a quantum model separates decoherence from record standing; whether a cosmological model states the boundary and witness ledgers by which its arrow is identified; and whether proposed new arrows remain effective projections rather than disguised primitives.

Part IX

Appendices

A Source-dependency index

This index records source dependence in prose rather than as a compressed table, so that the claim type and boundary of each source remain legible.

- **META.** Claim type: mathematical constraint layer. Function: irreversible commitment under unrestricted composition requires an admissibility invariant. Boundary: no physical ontology.

- **KERNEL.** Claim type: necessity-instantiated upstream theorem family. Function: proves that every non-degenerate regime of admissibility-assessable, identity-preserving construction already instantiates $K_D = (\text{Adm}_D, \text{Stand}_D, \text{Ref}_D, \text{Irr}_D)$, blocks faithful lower generation, and fixes the no-repair dependency. Boundary: no temporal primitive and no derivation of the kernel from the arrow theorem.
- **AMetric/ATS.** Claim type: interface discipline. Function: blocks primitive metricity, coordinates, scale, probability, and representative privilege. Boundary: no field equations or Standard Model content.
- **R1-R12 realized physics.** Claim type: role-source family. Function: supplies non-nullity, locality, domains, comparison, transport, fields, particles, gauge, mass, vacuum, time, and causality. Boundary: source status is declared explicitly and is not promoted beyond the established role scope.
- **R11 Time and Causality.** Claim type: direct theorem source. Function: supplies temporal-causal role induction, no primitive temporal parameter, record fixation, and causal role induction. Boundary: component theorem, not a full arrow-of-time synthesis.
- **Paper-level quotient theorem.** Claim type: main theorem discipline. Function: supplies the standing preorder on raw stages, quotient partial order on standing-equivalence classes, and strict standing arrow. Boundary: no new ontology.
- **R12 Realized Physical Interior.** Claim type: synthesis source. Function: supplies I_D , coherence, closure, and operational minimality. Boundary: not a full arrow-of-time paper.
- **Record construction / quantum records.** Claim type: companion theorem family. Function: supplies measurement-record quotient and record-use status. Boundary: no primitive collapse interpretation.
- **Bell witness arc.** Claim type: companion theorem family. Function: supplies joint standing compatibility classification. Boundary: no same-scope signalling arrow.
- **ZPE/vacuum arc.** Claim type: companion theorem family. Function: supplies standing discipline for absolute offsets and relational vacuum quantities. Boundary: no dark-energy or naturalness solution.
- **Cosmology/horizon arcs.** Claim type: support family. Function: supplies boundary, horizon, reconstruction, and witness-availability controls. Boundary: no full cosmological model.
- **The Physical Clock as a Quotient.** Claim type: downstream companion and Paper II of the time non-primitivity sequence; DOI: 10.5281/zenodo.20833048. Function: specializes the no-primitive-time discipline to physical clock standing, internal time, clock choice, and constrained quantum-gravity clock-use packages. Boundary: it is not imported as a premise of the arrow theorem; it is cited only to locate the sequel.

B Symbol index

Symbol	Meaning
K_D	AASC kernel: admissibility, standing, reference, irreversibility.
I_D	Realized physical interior.
$\mathcal{E}_D$	Raw class of standing-bearing stages in scope D .
$\preceq_S$	Standing preorder on raw stages.
$\equiv_S$	Standing equivalence: mutual standing precedence.
$\mathcal{E}_D / \equiv_S$	Standing quotient.
$\leq_S$	Partial order on standing-equivalence classes.
$<_S$	Strict standing arrow on the quotient.
T_D	Physical temporal role: ordered retained consequence.
$Q_{\text{Clk}}(D, U)$	Use-relative physical clock quotient of certified same-scope clock-use behavior; treated in Paper II.
Cau_D	Causal-dependence role: witnessed dependence across T_D .
R_D	Record-fixation quotient and record-reuse layer.
A_{therm}	Thermodynamic effective arrow.
A_{rec}	Record effective arrow.
A_{meas}	Quantum measurement-record arrow.
A_{cos}	Cosmological effective arrow.
A_{psych}	Psychological or agent-level projection arrow.
$\mathcal{A}_D$	Full arrow architecture.

C Glossary of core terms

Term	Meaning in this manuscript
Admissibility	The licensed status of a construction, description, or continuation inside a fixed scope.
Standing	Reusable status of admitted content under later lawful use.
Reference	The stability of what a claim, record, or comparison is about under admissible redescription.
Irreversibility	Non-erasure of standing-bearing consequence under same-scope admissible redescription; not primitive temporal flow.
Standing monotonicity	The non-invertible orientation induced by irreversible standing fixation.
Standing preorder	Raw-stage relation of standing retention, inheritance, conditioning, or transport.

Term	Meaning in this manuscript
Standing equivalence	Mutual standing precedence; the quotient identifies stages with no same-scope strict standing residue between them.
Standing arrow	The strict direction on standing-equivalence classes after quotienting.
Clock standing	Use-relative eligibility of a reference carrier for licensed physical readout, ordering, conditioning, or synchronization; not temporal direction.
Physical clock quotient	The canonical quotient $Q_{\text{CLK}}(D, U)$ of certified same-scope clock-use behavior for a declared domain and use; treated in Paper II.
Effective arrow	A domain-specific monotone, refinement, or projection admitted after its prerequisites have standing.
Primitive arrow	A directionality relation used as a foundation before its own standing conditions are fixed.
Record quotient	The equivalence class of record assignments that preserve the same standing-bearing use profile.
No-backflow	The rule that a downstream effective arrow cannot authorize the standing conditions required to identify it.
Anchor	A scope-setting condition such as a boundary, domain, regularity envelope, or comparison class.
Tensor	Standing-bearing content inside a fixed scope.
Skin	Presentation surplus: notation, coordinate, gauge representative, branch label, or display convention with no independent standing-bearing load.
Hidden load	Undeclared tensor or anchor work smuggled into a claim.
Redescription	Lawful change of presentation that preserves standing-bearing content.
Scope change	A legitimate change of theorem object or model domain, not a same-scope alternative.

D Arrow taxonomy table

Arrow type	Primitive?	AASC basis	Failure if primitive	Admissible survivor
Standing arrow	Forced, not a primitive source	Strict quotient order $<_S$	Not applicable	Invariant admissibility direction

Arrow type	Primitive?	AASC basis	Failure if primitive	Admissible survivor
Record arrow	No	Fixed witnesses and record quotient	Treats prediction as record	Evidence and memory arrow
Thermodynamic arrow	No	Admissible coarse-graining and entropy ledgers	Entropy backflow	Effective entropy monotone
Causal arrow	No	Witnessed dependence across T_D	Hidden influence or bare succession	Causal refinement
Quantum measurement arrow	No	Record-fixation quotient over predictive dynamics	Hidden selector or undeclared collapse	Measurement-record arrow
Psychological arrow	No	Agent-level memory and anticipation	Memory as primitive	Projection of record asymmetry
Cosmological arrow	No	Witness availability, constraint reachability, declared boundary anchors	Initial-condition backflow	Large-scale effective arrow
Information arrow	No	Encoding over stable reference	Encoding backflow	Post-admissible measure
Complexity arrow	No	Description scheme and comparison ledger	Information backflow	Effective complexity monotone

E Entropy interface ledger

Entropy object	Required ledgers	AASC status
Boltzmann $S_B = k_B \log \mu(\Gamma_M)$	State space, macroregion, measure, coarse-graining, macro/micro standing, comparison	Effective thermodynamic monotone
Gibbs $S_G = -k_B \int \rho \log \rho d\Gamma$	Probability density, measure/reference-density convention, state identity, normalization, ensemble interpretation	Post-admissible statistical entropy

Entropy object	Required ledgers	AASC status
Von Neumann $S(\rho) = -\text{Tr}(\rho \log \rho)$	Hilbert state standing, trace, density operator, subsystem/algebra ledger, record use if empirical	Quantum-statistical effective entropy

F Route-exhaustion normal forms

Normal form	Arrow-paper expression
Skin	Coordinate, label, branch name, or display convention with no standing-bearing content.
Hidden load	Undeclared entropy, probability, information, causal, or selector authority.
Anchor motion	New boundary condition, cosmological scope, state space, metric envelope, or admissibility precondition.
Tensor import	Declared new dynamics, state variable, field, collapse rule, hidden variable, or source ledger.
Quotient motion	Change to record equivalence, physical identity, or macro/micro quotient.
Continuation motion	Change to lawful update, dynamical rule, or admissible transition envelope.
Hidden selector	Branch, record, initial state, or outcome chosen without witness or declared scope.
Repair	Retroactive alteration of fixed record or same-act erasure of failure.
Metric backflow	Coordinate, proper time, foliation, clock, or light cone used before comparison is licensed.
Entropy backflow	Entropy used before state space, coarse-graining, and order have standing.
Information backflow	Encoding or complexity used before reference and redescription stability are licensed.

G Deletion matrix

Role	Delete it and...	Typical false survivor
Admissibility	There is no licensed/unlicensed distinction.	All arrows become descriptive preferences.
Standing	Fixed status cannot be reused.	Entropy or records become labels.
Reference	Ordered relata drift under redescription.	Coordinate sequences masquerade as time.
Irreversibility	No retained consequence distinguishes commitment from reversible exploration.	Formal dynamics replaces fixation.
Record quotient	Evidence and memory cannot be distinguished from prediction.	Information storage is treated as primitive.

Role	Delete it and...	Typical false survivor
Coarse-graining discipline	Entropy may erase tensor content or track skin.	Macrostate choice becomes hidden selector.
Witness dependence	Cause/effect becomes succession or influence without evidence.	Causal graph is promoted to primitive.
AMetric boundary	Coordinates, metrics, scale, or probability act pre-admissibly.	Metric time grounds itself.
No-backflow	Effective structures authorize their own prerequisites.	Every arrow can claim primitive status.

H Extension-claim protocol

Any extension of the arrow architecture should satisfy the following protocol.

1. Is the scope fixed?
2. Are anchors, tensors, and skin declared?
3. Does the proposed arrow preserve standing under lawful redescription?
4. Does it require record fixation, coarse-graining, witness dependence, or boundary data?
5. If so, are those roles imported explicitly rather than smuggled?
6. Is the arrow primitive, effective, representative, or merely skin?
7. Does the claim require a richer scope?
8. Does the route introduce repair, hidden selector, or metric/entropy/information backflow?
9. Does the arrow need global alignment with other arrows, or only local/domain alignment?
10. Is the counterexample burden stated for the relevant target?

I Claim-class ledger

Claim class	Examples in this paper	Control rule
Imported theorem family	Kernel, AMetric, RPI, record quotient	Use only at declared source scope; no backflow.
Paper-level theorem	Standing monotonicity, standing arrow uniqueness, plural effective arrows	Must be proved from imported roles and boundary eliminations.
Effective projection	Thermodynamic, causal, measurement, cosmological arrows	Declare quotient, witness, or boundary role.
Representative	Coordinate or proper time, metric cones, and branch labels	Do not promote to primitive standing.
Boundary anchor	Past Hypothesis, two-boundary cosmology, declared initial data	Must be declared as scope-setting, not pre-admissible.
Richer-scope tensor	Collapse dynamics, hidden variables, extra fields, nonlocal dynamics	Legitimate only with explicit tensor and witness ledger.

Claim class	Examples in this paper	Control rule
Theorem boundary	Numerical entropy law derivation, physical clock certification, multiple-choice closure, full cosmology, full quantum gravity	Assign to the declared downstream or richer scope.

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